



Herbal therapies for pain management: a scoping review of the current evidence

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Abstract Pain is a common symptom which can result in disability and lower quality of life. The current review covers the use of medicinal plants as

an alternative therapy for pain relief, as traditional painkillers like NSAIDs, opioids, and antidepressants can have serious side effects. Medicinal plants are effective, easily available, low-cost, and have fewer side effects. The review examines commonly used medicinal plants, their active components, their pharmacological activity, and their mechanism of

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action for different types of pain in humans and animal models. The review also discusses the use of herbal therapies for pain in various conditions, such as rheumatoid arthritis, neuropathies, osteoarthritis, dysmenorrhea, headache, migraine, wounds, low back pain, and chest pain, and weighs the advantages and disadvantages of using herbal therapies in light of recent research.

Keywords Pain management · Analgesic herbs · Alternative medicine · Complementary therapies · Phytomedicine · Herbal pharmacology

Abbreviations

NSAIDs	Non-steroidal anti-inflammatory drugs
IASP	International association for the study of pain
IBS	Irritable bowel syndrome
PRISMA-ScR	Preferred reporting items for systematic reviews and meta-analyses extension for scoping reviews
PICOS	Population, intervention, comparison, outcome, study design
PRISMA	Preferred reporting items for systematic reviews and meta-analyses
RA	Rheumatoid arthritis
IL	Interleukin
TNF	Tumor necrosis factor
MMPs	Matrix metalloproteinases
PGE2	Prostaglandin E2
COX-2	Cyclooxygenase-2
DMARDs	Disease-modifying anti-rheumatic drugs
NO	Nitric oxide
NF-κB	Nuclear factor kappa B
MAPK	Mitogen-activated protein kinase
OA	Osteoarthritis
BMI	Body mass index
ICAM-1	Intercellular adhesion molecule-1
iNOS	Inducible nitric oxide synthase
PI3K	Phosphoinositol 3 kinase
TLR	Toll-like receptor
LPS	Lipopolysaccharide
ROS	Reactive oxygen species
SOD	Superoxide dismutase
CFA	Complete Freund's adjuvant
IKB	Inhibitor of NF-κB

LOX	Lipoxygenase
hs-CRP	High sensitivity C-reactive protein
VAS	Visual analog scale
LBP	Low back pain
NMDA	N-methyl-D-aspartate
cAMP	Cyclic adenosine monophosphate
LPA	Lysophosphatidic acid
NGF	Nerve growth factor
SNRIs	Serotonin and non-adrenaline reuptake inhibitors
TCAs	Tricyclic antidepressants
CNS	Central nervous system
5-HT1A	5-Hydroxytryptamine receptor 1A
GABA	Gamma-aminobutyric acid
TRPA-1	Transient receptor potential ankyrin 1
KATP channels	ATP-sensitive potassium channels
TTH	Tension-type headaches
TXB2	Thromboxane B2
PAF	Platelet-activating factor
TRPM8	Transient receptor potential melastatin 8
THC	Tetrahydrocannabinol
TRPV1	Transient receptor potential vanilloid 1
EMS	Emergency medical services
ED	Emergency departments
STAT-3	Signal transducer and activator of transcription-3
ECGS-T segment	Electrocardiography ST segment
CAT	Chloramphenicol acetyltransferase
GSH	Glutathione
CBD	Cannabidiol
FDA	Food and drug administration
SJW	St. Johns wort
P gp	P Glycoprotein
RCTs	Randomized controlled trials
HDF	Human dermal fibroblasts
CCI	Chronic critical illness
Iba1	Ionized calcium-binding adaptor molecule 1
GFAP	Glial fibrillary acidic protein
PDGF	Platelet-derived growth factor
SPE	Solid phase extraction

Introduction

Pain is described as “An unpleasant sensory and emotional experience connected with actual or potential tissue damage, or described in terms of such damage,” according to the International Association for the Study of Pain (IASP) (Lewis 2003). Pain is one of the fundamental human health problems and can lead to significant suffering and disability (Vos et al. 2012) and remains one of the significant reasons people visit physicians. It is associated with many medical conditions created by intense or damaging stimuli, which lowers quality of life and puts a financial strain on healthcare (Williams and Craig 2016).

Generally, the sensation of pain is transmitted by afferent neurons from the periphery to the spinal cord (Peirs and Seal 2016). Although pain is a response to tissue injury, it can sometimes become chronic, leading to biological changes to the central nervous system or peripheral tissue. Additionally, pain can give rise to fear, anxiety, and increased hypothalamic stress response through cortical stimulation that eventually increases blood viscosity and platelet accumulation (Ahmad and Zakaria 2015). Every year, the prevalence of pain rises, especially among the elderly, as a result of longer life expectancies (Scurrah et al. 2018). The prevalence of pain is two times higher in older persons than in younger people, especially those over 60 years (Middaugh and Pawlick 2002). More than 100 million Americans are thought to be impacted by pain (Simon 2012), and 25% of Americans report experiencing pain every day (Gedin et al. 2017).

Pain can be neuropathic, visceral, visceral-chronic, inflammatory, or acute. Neuropathic pain is brought on by damage to the neurological system, whereas inflammatory pain is connected to tissue damage and immune cell infiltration (Pranay et al. 2023; Sommer 2016; Woolf 2010). Although there are numerous ways to categorize pain, it is frequently done so using time. Traditionally, pain can be classified as either being acute or chronic.

Acute pain typically results from damage or injury, lasts less than three months, and has no negative effects on the patient's quality of life. When pain is acute, its source and location are obvious. Acute pain (Apkarian et al. 2011) brought on by improper care can manifest as increased sympathetic nervous

system activity, tachycardia, tachypnea, etc. Trauma and surgical procedures are two instances of acute pain (Noroozian et al. 2018). In contrast, chronic pain might persist for three to six months or longer. The likelihood of acquiring chronic pain increases with age, previous pain experience, anxiety, stress, and depression, with females more likely to experience it (Apkarian et al. 2011). Mild chronic pain can occasionally become acute, episodic, or ongoing.

Sometimes mild to severe, episodic or persistent chronic pain might develop. It is a nociceptive or neuropathic pain-related public health issue (Noroozian et al. 2018). By activating somatic or visceral pain receptors, nociceptive pain is felt. Fractures of the bones and illnesses of the joints are a few examples of somatic discomfort. Irritable bowel syndrome (IBS), gastritis, and visceral discomfort are gastrointestinal conditions that cause pain. Neuropathic pain develops as a result of nerve injury brought on by radiotherapy or surgery (Noroozian et al. 2018).

Analgesic medications work differently to relieve pain by affecting both the peripheral and central neural systems. Typically, narcotics (opioids) and non-narcotics are used as analgesics (non-opioids). Pain treatment techniques typically involve the use of opiates like morphine, codeine, heroin, oxycodone, etc. and non-steroidal anti-inflammatory medicines (NSAIDs) such salicylates, COX-2 inhibitors, ibuprofen, etc. NSAIDs that are anti-inflammatory and analgesic are excellent treatments for mild to moderate pain. Opioid drug use, however, might result in adverse consequences such nausea, vomiting, respiratory depression, constipation, physical dependence, tolerance, etc. (Martyn et al. 2019). In recent years, medicinal plants have become more and more popular as a form of pain management due to the potential negative side effects and ineffectiveness of chemical medications (Bahmani et al. 2014).

Numerous studies have demonstrated the value of several herbal plants for treating pain (Bahmani et al. 2012; Bahmani and Eftekhari 2012; Jahromi et al. 2021; Kopustinskiene et al. 2022; Pranay et al. 2023; Song et al. 2023). Herbal remedies have historically been utilized as analgesic treatments due to their easy accessibility, low cost, little side effects, and potential to improve quality of life (Ayaz et al. 2016; Narwade et al. 2023; Ullah et al. 2015). Nearly 80% of people worldwide rely on plant-based medicines, according

to the World Health Organization (WHO) (Bahmani et al. 2014). Approximately 20 million Americans use herbal medicines in the United States (Bent 2008). Alkaloids, flavonoids, terpenoids, and steroids are among the active phytoconstituents extracted from these plants that inhibit the COX and LOX pathways, which have an impact on eicosanoid metabolism (Weiner and Ernst 2004). When taken in the right dosages, the active components of these plants have analgesic and anti-inflammatory properties, which can effectively relieve pain and have a positive impact on the course of the disease (Akhtar et al. 2011a). The main purpose of this scoping review is to evaluate the effectiveness and safety of medicinal plants in treating various pain related conditions. By examining both animal studies our aim is to understand the mechanisms through which these herbal remedies provide pain relief and reduce inflammation. Moreover we strive to contribute to the body of evidence based knowledge on pain management by synthesizing existing research findings. In essence, our goal is to provide healthcare professionals and researchers with an well informed perspective on the use of plants, for treating pain.

Materials and methods

An exploratory review design was employed to examine several medicinal plants' safety, efficacy, and potential uses for treating rheumatoid arthritis, osteoarthritis, dysmenorrhea, headaches, migraines, wounds, low back pain, and chest pain. We conducted this study based on the Arksey and O'Malley framework (Arksey and O'Malley 2007) and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for scoping reviews (PRISMA-ScR) (Tricco et al. 2018). In this exploratory review, five steps were followed: (i) Defining the eligibility criteria; (ii) identifying relevant studies through search strategy; (iii) selecting studies; (iv) extracting data; and (v) compiling and summarizing the findings.

Eligibility criteria

For this scoping review we have reported based on the PICOS (Population, Intervention, Comparison, Outcome, Study Design) methodology (Higgins et al. 2008). We have reported the findings utilising PRISMA-ScR methodology (Tricco et al. 2018).

Inclusion criteria

We will consider studies that involve both animal and human subjects that experience types of pain such as rheumatoid arthritis, neuropathic pain, osteoarthritis, dysmenorrhea, headache, migraine, wounds, low back pain or chest pain. There are no restrictions based on age or gender. Our focus will be on research that examines the use of plants for managing pain. We aim to include information on plant details including compounds and administration methods as well as outcomes related to pain relief. Studies comparing interventions will also be included. We will only consider research articles published in English that follow the PRISMA guidelines and can be both qualitative and quantitative.

Exclusion criteria

Our review excludes studies that do not involve animal or human subjects experiencing the specified types of pain. Additionally, studies that do not explore plant interventions for pain management or lack details about the intervention will not be included. Irrelevant studies unrelated to plant use for managing pain will also be excluded. Furthermore, studies where pain relief is not the primary focus will not be considered. We will exclude research articles that are not original publication in languages other than English.

Search strategy

A medical librarian (LÖ) conducted a systematic literature search in the four electronic databases: PubMed, EMBASE, Scopus and Web of Science. The search was conducted from time of inception to October 2022. Search terms relevant to the research question were systematically developed after discussing with subject matter experts. The search included PubMed and PubMeds' MeSH for the best possible search outcome. A combination of the search fields title, abstract and MeSH/Thesaurus and relevant truncation were applied to all search terms for the best possible search results: ((herb[Title/Abstract] OR herbs[Title/Abstract] OR herbal[Title/Abstract] OR herbals[Title/Abstract] OR "Herbal Medicine"[Mesh] OR "Plants, Medicinal"[Mesh] OR "Medicinal Plant*"[Title/Abstract] OR "Medical Plant*"[Title/Abstract] OR "Pharmaceutical Plant*"[Title/Abstract]

OR “Healing Plant”[Title/Abstract]) AND (“Pain”[-Mesh:NoExp] OR “Neuralgia”[Mesh] OR “Arthritis, Rheumatoid”[Mesh] OR “Osteoarthritis”[Mesh] OR “Dysmenorrhea”[Mesh] OR “Low Back Pain”[Mesh] OR “Migraine Disorders”[Mesh] OR “Headache”[Mesh] OR “Chest Pain”[Mesh] OR “Pain”[Title/Abstract] OR “Osteoarthritis”[Title/Abstract] OR “Dysmenorrhea”[Title/Abstract] OR “Low Back Pain”[Title/Abstract] OR “Migraine”[Title/Abstract] OR “Headache”[Title/Abstract] OR “Chest Pain”[-Title/Abstract] OR Neuralgia*[Title/Abstract] OR Neurodynia*[Title/Abstract] OR “Rheumatoid Arthritis”[Title/Abstract] OR Osteoarthritis[Title/Abstract] OR Osteoarthros*[Title/Abstract] OR Arthritis[Title/Abstract] OR Arthritides[Title/Abstract] OR Arthros*[Title/Abstract] OR dysmenorrhea*[Title/Abstract] OR Ache[Title/Abstract] OR Aches[Title/Abstract]. A detailed search can be found in online Appendix Table 1.

Data extraction

From the article, we extracted the following information: Scientific name, common name, family name, disease treated by the plant, bioactive phytochemicals, part of the plant used for treatment, and mode of administration. Additionally, we extracted scientific names for the plants, bioactive compounds, mechanisms of action of bioactive compounds, extracts that had an effect, extract dose, and human or experimental animal models with pain as outcomes (Table 1, Fig. 2).

Study selection and characteristics

The results of the search strategy are shown in Fig. 1. We include 378 published articles in the write-up. We have used a thematic analysis of the disease, the pathological disease process, extract dosage and the pain response as outcomes.

Rheumatoid arthritis

Rheumatoid Arthritis (RA) is a multifaceted autoimmune disease known for its progressive joint inflammation and a range of systemic complications, including extra-articular manifestations like rheumatoid nodules, pericarditis, and more (Cojocaru et al. 2010; Marcucci et al. 2018). This debilitating

condition affects approximately 1% of the global population, with a higher prevalence in North America and Europe, and a notable predilection for women (Gibofsky 2012; Köhler et al. 2019).

The etiology of RA is complex, involving genetic, hormonal, environmental, and immune factors. Factors such as smoking have been strongly associated with an increased risk of RA, with smokers being three to four times more likely to develop the disease (Lahiri et al. 2012; Sparks and Karlson 2016). Immune dysregulation plays a central role in RA, with activated CD4+T cells producing pro-inflammatory cytokines like interleukin IL-1, IL-6, and tumor necrosis factor (TNF)-alpha, leading to autoantibody production and joint damage (Choy and Panayi 2001; Mateen et al. 2016).

In the pathogenesis of RA, joint-damaging cytokines contribute to cartilage degradation by activating proteolytic enzymes like MMPs and collagenases (Jeong et al. 2016a, b; Moelants et al. 2013). Additionally, the activation of the p38 kinase signaling pathway by inflammatory cytokines further exacerbates joint pain and cartilage destruction (Ding et al. 2010; Ma et al. 2019). Arachidonic acid metabolism, specifically the conversion of arachidonic acid into prostaglandin E2 (PGE2) by COX-2 enzymes, plays a crucial role in the formation of pannus in synovial joints, contributing to synovial fibroblast apoptosis and joint inflammation (Amin et al. 2017; Farzaei et al. 2016; Linden et al. 2009).

Treatment strategies for RA encompass a spectrum of medications, including NSAIDs, immunosuppressants, pain relievers, disease-modifying anti-rheumatic drugs (DMARDs), and anti-TNF alpha agents, all aimed at alleviating pain, swelling, and joint damage (Hochberg et al. 2012; Hyrich et al. 2006; Soliman et al. 2012). However, long-term use of NSAIDs can lead to toxic side effects, and TNF-alpha inhibitors may induce hypersensitivity reactions and autoimmune responses (Cabral et al. 2016; McAlindon et al. 2014).

As a result, alternative therapies utilizing medicinal plant extracts have gained attention for their potential benefits in managing RA-related pain and inflammation pain (Table 1, Fig. 2, online Appendix Fig. 1).

1. *Arnica Montana*, a member of the Asteraceae family, has a long history of traditional use in treating RA and inflammation-related

Table 1 Plant based interventions for Rheumatoid arthritis

Plant (Scientific name)	Bioactive compounds	Mechanism of action	Extract	Dose	Experimental model	Results
1. <i>Arnica montana</i>	1,8 cineol, linalool, alpha-cadinol (Kowalski et al. 2015; Pljevljakušić et al. 2012; Sugier et al. 2017; Weremczuk-Jeżyna et al. 2011)	Anti-inflammatory Inhibits the cytokine induced airway mucus hypersecretion effectively	Methanolic extract (Sharma et al. 2016)	75 mg/kg body weight (Sharma et al. 2016)	Rat (Sharma et al. 2016)	Lessens histological and radiological changes in affected joints, reduces NO, IL-6, IL-12, IL-beta, TNF-alpha concentrations (Sharma et al. 2016)
2. <i>Curcuma spp.</i>	Curcumin (Nonose et al. 2014)	Improves inflammatory conditions in joint (Nonose et al. 2014)	Essential oil extract (Funk et al. 2010)	56 mg/kg/day (Funk et al. 2010)	Rat (Funk et al. 2010)	Inhibits joint swelling and arthritis induced by streptococcal cell wall (Funk et al. 2010)
3. <i>Equisetum arvense</i>	Kynurenic acid (Zgrajka et al. 2013)	Inhibits synoviocytes proliferation (Parada-Turska et al. 2006)	Root	15.89 pmol/ml	Human (Parada-Turska et al. 2006)	Inhibits proliferation of synoviocytesml
4. <i>Salix spp.</i>	Salicin (Bonaterra et al. 2010; Drummond et al. 2013; Khayyal et al. 2005; Shara and Stohs 2015)	Decreases proinflammatory mediators and inhibits COX 2 activity (Bonaterra et al. 2010; Drummond et al. 2013; Khayyal et al. 2005; Shara and Stohs 2015)	Bark extract (Biegert et al. 2004)	240 mg/day (Biegert et al. 2004)	Human (Biegert et al. 2004)	Relieves pain of Ra (Biegert et al. 2004)
5. <i>Aloe barbadensis</i>	Antraquinone (Sharma et al. 2014a, b)	Inhibits PGE2 and iNO (Budai et al. 2013; Kshirsagar et al. 2014)	Methanolic extract (Kshirsagar et al. 2014)	25, 50, and 75 mg/kg of aloe emodin (Kshirsagar et al. 2014)	Rat (Kshirsagar et al. 2014)	Inhibits PGE2, IL-8, IL-6, IL-3, TNF-alpha production and translocation of NF-KB kinases, p ³⁸ , JNK and ERK
6. <i>Andrographis paniculata</i>	Andrographolide (Gupta et al. 2017; Li et al. 2017a, b)	Reduces PGE2 production (Wang et al. 2004; Yan et al. 2012)	Ethanollic, Dried	10, 20, and 30 µM of	Human (Yan et al. 2012)	andrographolide (Yan et al. 2012)
7. <i>Berberis orthobotrys</i>	Beta-sitosterol (Mehmood et al. 2018)	Inhibits NF-KB and regulates genes for increased inflammatory mediators (Valerio and Awad 2011)	Aqueous methanolic extract (Alamgeer et al. 2017)	150 mg/kg (Alamgeer et al. 2017)	Rat (Alamgeer et al. 2017)	Exhibits anti-arthritic activity by inhibiting IFN-alpha, PDGF, IL-1, IL-6, TNF-alpha etc. cytokine production (Alamgeer et al. 2017)
8. <i>Celastrus paniculatus</i>	Celastrine, celapagine, beta-sitosterol, beta-amyrin (Shashank et al. 2017; Shimizu et al. 2015)	Inhibits TNF-alpha and IL-6 (Kulkarni et al. 2015)	Ether seed extract (Kulkarni et al. 2015)	(500 and 400) mg/kg (Kulkarni et al. 2015)	Mice (Kulkarni et al. 2015)	Inhibits TNF-alpha, IL-6 and inflammatory mediators involved in swelling, joint devastation (Kulkarni et al. 2015)
9. <i>Oenothera biennis</i>	Gamma linolenic acid (Voukeng et al. 2016)	Reduces COX 2 gene expression and modulates NO, TNF-alpha, IL-1 beta, TXB2 pathway (Voukeng et al. 2016)	Seed oil	1.4 g/day	Human	Exhibits antiinflammatory and immunoregulatory properties
10. <i>Uncaria tomentosa</i>	Pentacyclic oxindole, proanthocyanidins (Navarro et al. 2019)	Inhibits IL-1 alpha, IL-1 beta, IL-4, IL-17 and TNF-alpha (Navarro et al. 2019)	Krallendorn® capsule contains 20 mg of aqueous acid-extracted dry extract of U. tomentosa (Mur et al. 2002)	(2386.5 µg/g)	Human (Mur et al. 2002)	reduction of the number of painful joints (Mur et al. 2002)

Table 1 continued

Plant (Scientific name)	Bioactive compounds	Mechanism of action	Extract	Dose	Experimental model	Results
11. <i>Withania somnifera</i>	Withanolides, withaferin A (Khan et al. 2015; Singh et al. 2007)	Inhibits TNF-alpha, IL-1beta, IL-12, NF-KB (Ganesan et al. 2011; Grover et al. 2010; Heyninck et al. 2014; Singh et al. 2007)	Aqueous root extract (Ramakanth et al. 2016)	500 mg/day (Ramakanth et al. 2016)	Human (Ramakanth et al. 2016)	Reduces pain, stiffness and deformities in patients with knee joint pain (Ramakanth et al. 2016)
12. <i>Caragana pruniosa</i>	Purinosanone D and E, butin, scutellaprostin C and 2,4-dihydroxy flavanone (Meng et al. 2009; Peng et al. 2016; Sun et al. 2015; Zaka et al. 2017)	Downregulates TNF-alpha, IL-1beta, IL-6 and CRP (Meng et al. 2009; Peng et al. 2016; Sun et al. 2015)	Ethanollic root extract (Peng et al. 2016; Zhang et al. 2019)	(130, 260 and 520) mg/kg (Peng et al. 2016; Zhang et al. 2019)	Rat (Peng et al. 2016; Zhang et al. 2019)	Suppresses TNF-alpha, IL-1beta, IL-6, IL-10 and CRP levels elevated in RA (Peng et al. 2016; Zhang et al. 2019)
13. <i>Eysenhardtia polystachya</i>	Stigmasterol, isoduartin, cuneatin and 3,4-dimethoxy-8,9-pterocarpan (Alonso-Castro et al. 2018; Pablo-Pérez et al. 2018)	Reduces serum IL-6 and TNF-alpha etc. proinflammatory cytokines (Alonso-Castro et al. 2018; Pablo-Pérez et al. 2018)	Ethanollic extract (Pablo-Pérez et al. 2018)	(25,50,100 and 200) mg/kg/day (Pablo-Pérez et al. 2018)	Rat (Pablo-Pérez et al. 2018)	Reduces arthritic pain by decreasing serum IL-6, TNF-alpha, GM-CSF etc. proinflammatory cytokines production (Pablo-Pérez et al. 2018)
14. <i>Inula helenium</i>	Alantolactone, isoalantolactone (Chun et al. 2012; Gao et al. 2017)	Inhibits TNF-alpha, MCP1, MCP3, IL-1, IL-6, MCP2 expression (Younis et al. 2016)	Root extract (Gao et al. 2017)	(12.5–50) mg/kg (Gao et al. 2017)	Rat (Gao et al. 2017)	Reduces RA symptoms by inhibiting TNF-alpha induced NF-KB and MAPK pathway activation (Gao et al. 2017)
15. <i>Nigella sativa</i>	Thymoquinone (Umar et al. 2012; Vaillancourt et al. 2011)	Hinders IL-1beta, TNF-alpha, IL-10, IFN-gamma, PGE2, IL-6 expression (Umar et al. 2012)	Capsules containing <i>Nigella sativa</i> essential oil (Hadi et al. 2016)	1000 mg/day (Hadi et al. 2016)	Human (Hadi et al. 2016)	Improves inflammation by increasing IL-10 and reduces oxidative stress in RA (Hadi et al. 2016)
16. <i>Strychnos nux-vomica</i>	Brucine, Brucine-N-oxide (Yin et al. 2003)	Suppress PGE2 production and decrease vascular permeability (Chaurasia 2009; Ekambaram et al. 2010)	aqueous extract (SPE), whole seed powder	200 mg/kg/p.o (Ekambaram et al. 2010)	Rat model (Ekambaram et al. 2010)	Reduces paw edema volume (Ekambaram et al. 2010)

conditions. Its active compounds, including 1,8-cineol, linalool, and alpha-cadinol, have shown promise in reducing inflammation and improving joint health (Sharma et al. 2016). Studies on

a collagen-induced arthritic rat model demonstrated that phenolic and flavonoid compounds from Arnica Montana extracts could effectively mitigate histological and biochemical changes

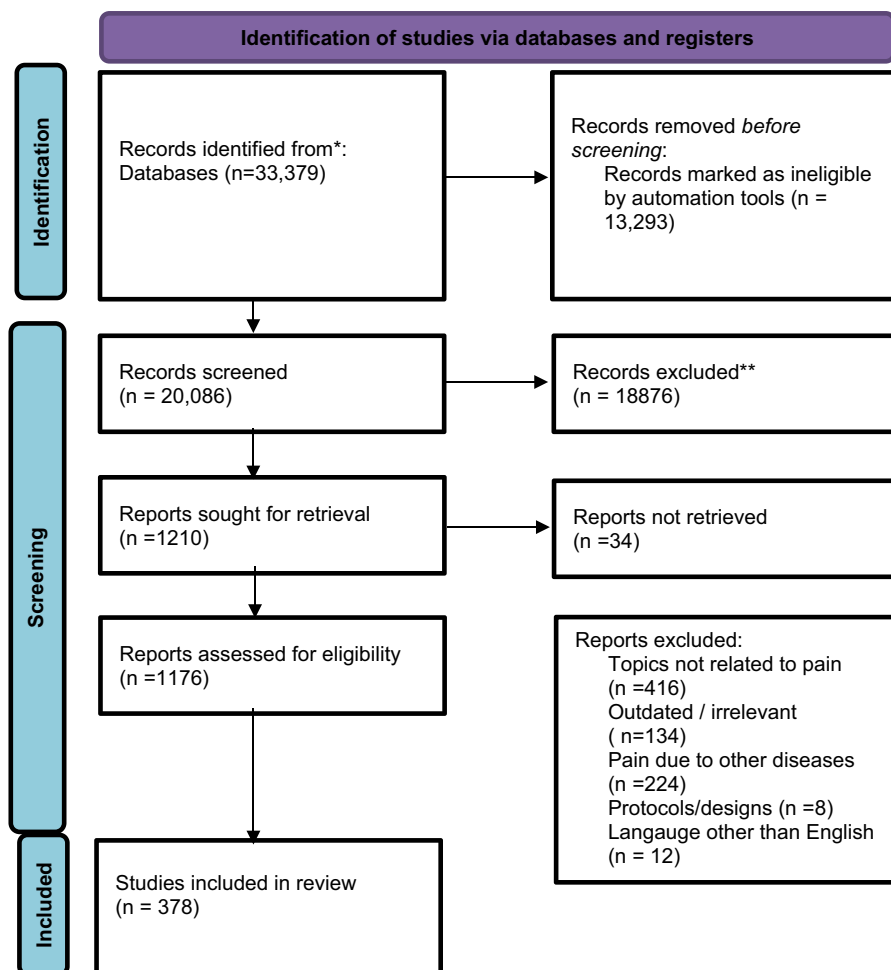


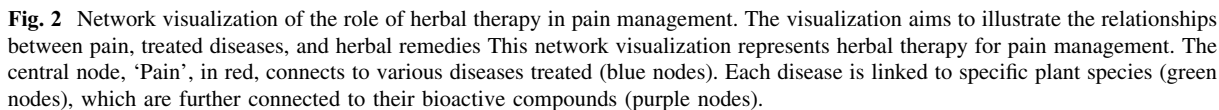
Fig. 1 PRISMA

in affected joints. These compounds also reduced levels of pro-inflammatory cytokines such as nitric oxide (NO), IL-6, IL-12, IL-1beta, and TNF-alpha, while enhancing antioxidant defenses and reducing peroxidative injury (Sharma et al. 2016).

2. *Curcuma species*, particularly *Curcuma longa* or turmeric, have gained popularity for their anti-inflammatory properties. Curcumin, a phenolic compound found in turmeric, has shown efficacy in suppressing joint damage and inflammation by inhibiting various inflammatory mediators (Prasad and Aggarwal 2011). It also prevents destructive changes in joints and bones (Gründemann et al. 2014; Li et al. 2013). Notably, curcumin exhibited superior efficacy

to prednisone in reducing neutrophil infiltration in rat models of zymosan-induced arthritis (Nonose et al. 2014). Coumarin, another active compound, contributes to systemic oxidative stress reduction (Panahi et al. 2016). Beta-element, derived from *Curcuma wenyujin*, has been used traditionally in China for RA treatment due to its anti-proliferative activity against fibroblast-like synoviocytes (Zou et al. 2016).

3. *Equisetum arvense*, commonly known as horsetail, has been employed in European ethnomedicine for its anti-inflammatory properties. Clinical trials have indicated its potential in reducing TNF-alpha levels in RA patients (Jiang et al. 2014). Kynurenic acid, an active



et al. 2010; Drummond et al. 2013; Khayyal et al. 2005). Additionally, willow bark extracts have been shown to block the activity of TNF-alpha and NF-kB, further supporting their anti-inflammatory properties (Bonaterra et al. 2010). Clinical trials have demonstrated the effectiveness of willow bark extract in relieving RA pain, with results comparable to the nonsteroidal anti-inflammatory drug (NSAID) diclofenac (Biegert et al. 2004).

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5. *Aloe barbadensis*, commonly known as aloe vera, contains various chemical constituents with anti-inflammatory properties, including anthraquinone alkaloids (Jales et al. 2021). These compounds have demonstrated effectiveness in treating arthritis and rheumatic conditions (Budai et al. 2013). Aloe vera extract has been found to inhibit the production of pro-inflammatory cytokines such as PGE2, IL-8, IL-6, IL-1, and TNF-alpha, as well as the activation of signaling pathways involved in inflammation (Yagi and Yu 2018).
6. *Andrographis paniculata*, an herbaceous plant of the Acanthaceae family, contains andrographolide and its derivatives, which exhibit anti-rheumatoid activity by inhibiting NF-kB activation (Gupta et al. 2017). Clinical trials have demonstrated the potential of andrographolide in reducing RA symptoms (Wang et al. 2004; Yan et al. 2012). This plant offers a potential avenue for RA management.
7. *Berberis arthrobotrys*, a member of the Berberidaceae family, contains active phytoconstituents such as beta-sitosterol that regulate genes associated with inflammation and inhibit NF-kB in rheumatoid arthritis (Uttra et al. 2019). Aqueous methanolic extracts of this plant have shown anti-arthritic activity by suppressing the production of pro-inflammatory cytokines in animal models (Valerio and Awad 2011). These findings suggest a potential role for *Berberis arthrobotrys* in RA treatment.
8. *Celastrus paniculatus*, locally known as black oil, possesses anti-inflammatory, antioxidant, and antipyretic properties due to compounds like sesquiterpenoids and alkaloids (Shashank et al. 2017). This plant has demonstrated the ability to inhibit pro-inflammatory mediators, including TNF-alpha and IL-6, in animal models (Kulkarni et al. 2015). It shows promise as a natural remedy for RA.
9. *Oenothera biennis*, or evening primrose, contains gamma-linolenic acid, which has been shown to reduce the pathogenesis of RA by modulating inflammatory mediators (Voukeng et al. 2016). This compound can suppress the expression of genes associated with inflammation, including COX-2, and reduce oxidative stress in RA (Voukeng et al. 2016).
10. *Uncaria tomentosa*, commonly known as cat's claw, contains active compounds like pentacyclic oxindole and proanthocyanidins with anti-inflammatory properties. It has been found to alleviate pro-inflammatory cytokines and improve RA symptoms in animal models (Sordi et al. 2019). Clinical trials have shown its potential in reducing joint pain in RA patients (Mur et al. 2002). *Uncaria tomentosa* may offer a natural approach to RA management.
11. *Withania somnifera*, known as ashwagandha, contains active compounds called withanolides, including Withaferin A, with antioxidant properties (Singh et al. 2011). These compounds inhibit the production of pro-inflammatory cytokines like TNF-alpha and IL-1 beta, as well as NF-kB activation (Singh et al. 2007). *Withania somnifera* has shown promise in reducing pain, stiffness, and deformities in RA patients (Ramakanth et al. 2016).
12. *Caragana pruinosa*, a member of the Leguminosae family, contains active constituents like flavonoids and purinosanones that downregulate pro-inflammatory cytokines such as TNF-alpha and IL-6 (Zaka et al. 2017). Preclinical studies have demonstrated its effectiveness in improving RA symptoms (Peng et al. 2016; Sun et al. 2015).
13. *Eysenhardtia polystachya*, known as Palo Azul, has been used traditionally for its anti-inflammatory properties. Studies have shown that its ethanolic extract can reduce arthritic pain and levels of pro-inflammatory cytokines in animal models (Huang et al. 2014a, b). This plant extract may hold promise as a natural remedy for RA.
14. *Inula helenium*, a member of the Asteraceae family, contains active compounds like alantolactone and isoalantolactone, which inhibit TNF-alpha-induced NF-kB and MAPK pathway activation (Gao et al. 2017). *Inula helenium* has shown potential in reducing pro-inflammatory cytokines and may have a role in RA management (Younis et al. 2016).
15. *Nigella sativa*, known as kalonji or black caraway, contains thymoquinone, which inhibits the expression of pro-inflammatory cytokines like IL-1beta, TNF-alpha, and IL-6 (Umar et al. 2012; Vaillancourt et al. 2011). Clinical trials have demonstrated its effectiveness in reducing inflammation and oxidative stress in RA patients (Hadi et al. 2016).

Table 2 Plant based interventions for Osteoarthritis

Plant (Scientific name)	Bioactive compounds	Mechanism of action	Extract	Dose	Experimental model	Results
1. <i>Boswellia serrata</i>	11 keto boswellic acid and acetyl 11 keto beta boswellic acid (Hamidpour et al. 2013; Sharma et al. 1989; Wang et al. 2014)	Improves synovial fluid protein pattern and reduces the leukocyte infiltration (Hamidpour et al. 2013; Sharma et al. 1989; Wang et al. 2014)	Acetone extract (Wang et al. 2014)	34.14 mg/gKBA and 74.62 mg/g AKBA	Rat (Wang et al. 2014)	Improves arthritic conditions (Wang et al. 2014)
2. <i>Harpagophytum procumbens</i>	Harpagoside (Boje et al. 2003; Mncwangi et al. 2012)	Inhibits iNOS and COX2 expression (Huang et al. 2006)	Aqueous extract (Wegener and Lüpke 2003)	2400 mg/day (Wegener and Lüpke 2003)	Human (Wegener and Lüpke 2003)	Reduces knee and hip osteoarthritis pain (Wegener and Lüpke 2003)
3. <i>Eucommia ulmoides</i>	Aucubin (He et al. 2014a, b, c)	Decreases cell apoptosis, ROS production and caspase-3 activity (Young et al. 2017)	Aqueous extract (Lu et al. 2013)	6 ml/kg/day (Lu et al. 2013)	Rat (Lu et al. 2013)	Reduces the serum MMP-13, MMP-1, MMP-3 and protects articular cartilage (Lu et al. 2013)
4. <i>Cinnamomum cassia</i>	Cinnamaldehyde (Zhou et al. 2019)	Increases glutathione peroxidase, SOD and catalase activity (Zhou et al. 2019)	Hydroalcoholic bark extract (H. Sharma et al. 2018)	250 mg/kg (H. Sharma et al. 2018)	Rat (H. Sharma et al. 2018)	Reduces formaldehyde and CFA induced arthritis (H. Sharma et al. 2018)
5. <i>Centella asiatica</i>	Madecassoside (Moqbel et al. 2020)	Suppresses NF-Kb activation pathway (Cao et al. 2010)	Nuclear extracts	300 µM	Rat	Prevention of hypotension, Protects cardiovascular and inflammatory agent
6. <i>Withania somnifera</i>	Withaferin A (Grover et al. 2010; Heyninck et al. 2014)	Suppresses NF-KB activation (Grover et al. 2010; Heyninck et al. 2014)	Aqueous root extract (Ramakanth et al. 2016)	(120 and 250)mg (Ramakanth et al. 2016)	Human (Ramakanth et al. 2016)	Reduces pain, disability, stiffness and knee swelling index (Ramakanth et al. 2016)
7. <i>Zingiber officinalis</i>	6-dehydro gingerdione (Ji et al. 2017; Schadich et al. 2016)	Alleviates iNOS, COX-2, IL-beta, IL-6 and TNF-alpha expression (S.-H. Huang et al. 2014a, b, p. 264)	Methanol	40 µg/m	Qiagen Kits	Activate NRF2 & expression of GSTP1

Table 2 continued

Plant (Scientific name)	Bioactive compounds	Mechanism of action	Extract	Dose	Experimental model	Results
8. <i>Sesamum indicum</i>	Sesamin, sesamol, sesamolin (Rangkadilok et al. 2010)	Inhibits LOX, TNF-alpha, IL-1 beta, IL-6, MMP-13 and MMP-9 activity (Hsu et al. 2013, 2016; Sotnikova et al. 2009)	Sesame seed powder (Haghighian et al. 2015)	40 g (Haghighian et al. 2015)	Human (Haghighian et al. 2015)	Reduces serum inflammation and oxidative stress in patients with knee OA (Haghighian et al. 2015)

16. *Strychnos nux-vomica*, commonly known as poison nut, contains brucine and brucine N-oxide, which have anti-inflammatory properties (Yin et al. 2003). Preclinical data suggests that *Strychnos nux-vomica* can suppress PGE2 and decrease vascular permeability, providing potential relief from arthritic pain (Chaurasia 2009; Ekambaram et al. 2010). These medicinal plant extracts and their active constituents offer promising alternatives for managing RA-related pain and inflammation, providing potential relief with fewer adverse effects than conventional medications. Table 1 summarizes the plant extracts and their mechanisms of action, which have shown promise as complementary therapies for RA (Table 1, Fig. 2).

Osteoarthritis

Osteoarthritis (OA) is one of the most common types of joint disorder which is associated with joint pain. OA is a long-term, chronic musculoskeletal disorder that affects 15% of the population (Hsu et al. 2016). Joint pain is a significant symptom of OA that reduces the patient's quality of life (Owens and Conaghan 2016), with the knee and back being the most commonly affected joints (Yudoh et al. 2017). Eighty percent of the human population suffering from OA are over 65 years old (Mahdavi et al. 2018). OA affects women more than men and is the leading cause of pain and disability in older people (Wang et al. 2015). The risk of developing osteoarthritis is increased by increased body mass index (BMI), older age, genetics, obesity, diet, and physical activities

that cause stress across joints (Murphy et al. 2016). Traumatic joint injuries, particularly knee injuries, can lead to a five–sixfold increase in the risk of developing OA (Lane et al. 2017). OA is characterized by bone erosion, increased pro-inflammatory cytokines including IL-1, IL-6, and TNF-alpha, and irreversible articular cartilage degeneration (Hsu et al. 2016), leading to increased synthesis of collagenase, degradation of matrix metalloproteinase (MMP), and decreased collagenase inhibitor synthesis (Park et al. 2009). Breakdown of the extracellular matrix stimulates the accumulation of innate immune cells leading to inflammation and tissue destruction (Mora et al. 2018). Signaling pathways such as mitogen-activated protein kinase (MAPK) play a role in OA (Mobasher & Batt 2016). IL-8, IL-17, IL-18, IL-21 diffuse into the synovial fluid, act on chondrocytes and impair matrix synthesis (Khalid et al. 2017). OA is associated with bone thickening, ligament degeneration, and synovial inflammation (Kolasinski et al. 2020). Both peripheral and central mechanisms contribute to OA pain. Peripheral pain is observed in the early stage of osteoarthritis, while central pain is observed in the late and chronic stages (Arendt-Nielsen et al. 2010). NSAIDs, acetaminophen, opioids, intraarticular injections are used in OA treatment. Acetaminophen, NSAIDs, and opioids inhibit the synthesis of prostaglandins and cyclooxygenase enzyme, which helps in pain relief. However, these drugs have adverse effects as well. Long-time use of NSAIDs is associated with nephrotoxicity gastrointestinal and cardiovascular effects (Solomon et al. 2018). Numerous herbs offer alternative approaches to alleviate the pain and symptoms associated with OA. Below, is a list of medicinal

plant extracts and their active phytoconstituents that are employed in the treatment of OA-related pain (Table 2, Fig. 2, online Appendix Fig. 2).

1. *Boswellia serrata*, a plant with a history in Ayurvedic medicine for treating arthritis, exhibits anti-inflammatory and cartilage-protective properties. In vitro studies revealed its ability to resist cartilage degradation by inhibiting MMP-3 and blocking ICAM-1 (Sengupta et al. 2010). Moreover, it hinders collagen and cartilage breakdown by reducing MMP-9, MMP-13, COX-2, NO, and PGE2 production (Blain et al. 2010). Boswellic acids, such as 11-keto boswellic acid and acetyl 11-keto-beta boswellic acid, improve synovial fluid proteins and reduce joint inflammation (Sharma et al. 1989; Wang et al. 2014). This herb also suppresses NF-kB activation and pro-inflammatory cytokines (Ammon 2010; Khajuria et al. 2008). Combining *Boswellia* with other herbs like *Zingiber officinale*, *Tinospora cordifolia*, and *Phyllanthus emblica* further protects cartilage (Sumantran et al. 2011). *Boswellia* extract enhances antioxidant status by reducing NO, myeloperoxidase, and lactoperoxidase (Umar et al. 2014). Additionally, a mixture of *Boswellia*, *Withania somnifera*, *Zingiber officinale*, and *Curcuma longa* alleviates arthritis symptoms by decreasing TNF-alpha and NO production (Dey et al. 2014). The combination of *Uncaria tomentosa*, *Boswellia serrata*, *Lepidium meyenii*, and leucine protects articular cartilage, inhibits NF-kB activation, and enhances aggrecan and type II collagen production in OA chondrocytes (Akhtar et al. 2011b). Phytopreparations from *Boswellia* resin have shown efficacy in relieving pain and improving functionality in knee and hand OA patients without severe toxicity (Belcaro et al. 2015). Furthermore, a combination of *Curcuma longa* and *Boswellia* appears more effective and safe in managing OA symptoms than the COX-2 inhibitor celecoxib (Kizhakkekath 2013).
2. *Harpagophytum procumbens*, commonly known as devil's claw, exhibits chondroprotective activity by reducing the release of pro-inflammatory cytokines like TNF-alpha and IL-1beta and inhibiting the synthesis of MMP and esterase (Fiebich et al. 2001). Clinical studies have supported the use of aqueous extracts of *Harpagophytum procumbens* (at 2400 mg/day) in reducing pain associated with knee and hip osteoarthritis (Chantre et al. 2000; Chrubasik et al. 2002; Wegener and Lüpke 2003). Harpagoside, a compound found in the plant, inhibits iNOS and COX-2 expression by blocking NF-kB activation (Huang et al. 2006).
3. *Eucommia ulmoides*, a member of the Eucommiaceae family, shows promise in osteoarthritis treatment. It reduces the production of pro-inflammatory cytokines such as IL-6, IL-17, TNF-alpha, and IL-1beta in an OA rat model by suppressing the phosphoinositol 3 kinase (PI3K)/Akt signaling pathway activation (Wang et al. 2015). In murine macrophages, *Eucommia ulmoides* modulates COX-2 and inhibits toll-like receptor (TLR)4, TNF-alpha, IL-1beta, iNOS, and IL-6 production induced by LPS (Koh et al. 2017). Aucubin, an active compound from this plant, reduces cell apoptosis, ROS production, and caspase-3 activity (Young et al. 2017). When used in combination with meloxicam, *Eucommia ulmoides* has demonstrated better pain relief compared to meloxicam monotherapy (Hu et al. 2020). Additionally, an aqueous extract (6 mg/kg/day) of *Eucommia ulmoides* protects articular cartilage by reducing serum MMP-13, MMP-1, and MMP-3 levels in an OA animal model (Lu et al. 2013).
4. *Cinnamomum cassia*, a traditional Chinese medicine, is employed for osteoarthritis treatment due to its anti-inflammatory effects and low toxicity (Cross et al. 2014). It influences apoptosis, cell proliferation, differentiation, and the balance of osteoblasts and osteoclasts (Hootman et al. 2016). Cinnamaldehyde, an active compound, inhibits chondrocyte oxidation by increasing the activity of enzymes like glutathione peroxidase, SOD, and catalase (Zhou et al. 2019). The hydroalcoholic bark extract of *Cinnamomum cassia*, at a dose of 250 mg/kg, reduces arthritis induced by formaldehyde and CFA (Sharma et al. 2018).
5. *Centella asiatica*, commonly known as 'Gotu kola,' is distributed widely in Asia and is utilized for its antioxidant and anti-inflammatory activities in OA treatment (Prakash et al. 2017). The plant contains various phytochemicals, including

- tannins, polyphenols, beta-carotene, vitamin C, and flavonoids, which contribute to its antioxidant activity (Chandrika and Prasad Kumara 2015). Madecassoside, a major phytoconstituent, suppresses the NF- κ B activation pathway and protects cartilage by inhibiting the effects of IL-1 β on bone marrow cells (Moqbel et al. 2020). *Centella asiatica* also inhibits NO production and exhibits disease-modifying activity in OA patients (Hartog et al. 2009). Oral administration of *Centella asiatica* attenuates collagen-induced arthritis by reducing NO and pro-inflammatory cytokine production (Sharma et al. 2014a, b).
6. *Withania somnifera*, known for its analgesic and anti-inflammatory properties in Ayurvedic medicine, inhibits NF- κ B activation, ROS production, and the secretion of pro-inflammatory cytokines such as IL-6, TNF- α , MMP-8, IL-1 β , and IL-10 (Khan et al. 2019). Withaferin A, an active ingredient, suppresses NF- κ B activation by reducing the I κ B association complex (Grover et al. 2010; Heyninck et al. 2014). *Withania somnifera* also inhibits collagenase activity and improves joint disease (Ganesan et al. 2011). In collagen-induced arthritic rats, it alleviates symptoms such as redness, ankylosis, swelling, and deformity (Gupta and Singh 2014). Compared to a placebo, a 250 mg aqueous root extract of *Withania somnifera* taken once daily reduces pain, disability, stiffness, and knee swelling index (Ramakanth et al. 2016).
 7. *Zingiber officinale*, known for its anti-inflammatory properties, inhibits COX-1, COX-2, and LOX in vitro (Huang et al. 2014a, b; Van Breemen et al. 2011). Human clinical studies have demonstrated that the administration of 1 g/day of *Zingiber officinale* powder for three months improves knee osteoarthritis by reducing serum nitric oxide, high sensitivity reactive protein (hs-CRP), and other inflammatory markers (Naderi et al. 2016). Other studies have supported its effectiveness in improving the clinical condition of OA patients as evaluated by the VAS scale (Alipour et al. 2017). In combination with glucosamine, *Zingiber officinale* effectively reduces pain in knee and hip OA patients (Drozdov et al. 2012). Active compounds such as 1-dehydrogingerdione and 6-dehydrogingerdione inhibit NF- κ B activation and reduce inflammatory gene expression (Lee et al. 2012; Yang et al. 2012).
 8. *Sesamum indicum*, commonly known as sesame, has been used in traditional Asian medicine for its anti-inflammatory properties. Sesame oil inhibits inflammation by blocking NF- κ B activation and reducing the levels of TNF- α , IL-1 β , and IL-6 in arthritic rats (Hsu et al. 2013). It also reduces joint pain by blocking lipid peroxidation and superoxide production (Hsu et al. 2016). Sesame seeds have been shown to lower serum inflammation markers in knee OA patients (Haghighian et al. 2015). Active phytochemicals in sesame, such as sesamin, sesamol, and sesamol, inhibit LOX, TNF- α , IL-1 β , IL-6, MMP-13, and MMP-9 activity (Hsu et al. 2013, 2016; Sotnikova et al. 2009). Sesamol has been found to reduce joint inflammation, cartilage degradation, and bone resorption in arthritic animals (Hemshekhkar et al. 2012).

In summary, these natural remedies offer promising avenues for osteoarthritis treatment, with potential benefits in reducing inflammation, protecting cartilage, and alleviating pain. However, further research and clinical trials are necessary to confirm their efficacy and safety in clinical practice.

Low back pain

Low back pain (LBP) is a prevalent public health issue, ranking as the second most common reason for physician visits, the fifth major cause of hospitalization, and the third most frequent indication for surgery (Goetzel et al. 2015). The consequences of LBP extend to limitations in daily activities, including the inability to work (Natour et al. 2014). Chronic LBP exacerbates individuals' suffering and escalates treatment costs (Liao et al. 2009). In response to this complex health issue, diverse treatment strategies have been employed, encompassing physical therapy, surgical interventions, NSAIDs, and early resumption of regular physical activity (Last and Hulbert 2009). In developed countries, additional approaches such as injections, opioid therapy, spinal imaging, and surgical fusions are employed in the management of LBP (Madden et al. 2013). Beyond conventional treatments, some plant-based remedies and their active components have gained attention for their efficacy, minimal side effects, and accessibility in LBP

Table 3 Plant based interventions for Low back pain

Plant (scientific name)	Bioactive compounds	Mechanism of action	Extract	Dose	Experimental model	Results
1. <i>Aloe barbadensis</i>	Acemannan, oleic acid (Salehi et al. 2018)	Inhibition to COX enzymes	Petroleum ether, water, upper steam, young bark, mature bark, Leaves & Roots, Acetone	1.0 ml/kg bodyweight, 1.56 mg/ml	Rats, Agar plate	Phagocytic and proliferative activity, Antimicrobial
2. <i>Salix alba</i>	Salicin (Bonaterra et al. 2010; Drummond et al. 2013; Khayyal et al. 2005; Shara & Stohs 2015)	Inhibits COX activity and reduces pro-inflammatory mediators synthesis (Bonaterra et al. 2010; Drummond et al. 2013; Khayyal et al. 2005; Shara & Stohs 2015)	Bark extract (Chrubasik et al. 2000)	(120 and 240) mg (Chrubasik et al. 2000)	Human (Chrubasik et al. 2000)	Reduces the pain intensity (Chrubasik et al. 2000)
3. <i>Zingiber officinale</i>	Curcumin, salicylate (Altman and Marcussen 2001)	inhibition of cyclooxygenase and lipooxygenase pathways (Lem and Lee 2017)	Ginger compress (Lem and Lee 2017)	255 mg	Human (Lem and Lee 2017)	Reduced symptoms of backpain
4. <i>Curcuma longa</i>	Curcumin (Abdel-Lateef et al. 2016; Altman & Marcussen 2001)	Inhibits COX and LOX synthesis, NF-KB, TNF- α etc. pro-inflammatory cytokines (Ramadan et al. 2010)	200 mg/kg body weight turmeric powder in 1.0 ml distilled water (Ramadan et al. 2010)	200 mg/kg (Ramadan et al. 2010)	Rat model (Ramadan et al. 2010)	Suppressed severity of arthritis (Ramadan et al. 2010)

management. Various plants and their active ingredients have gained attention in the management of low back pain due to their perceived efficacy, minimal side effects, and accessibility (Table 3, Fig. 2, online Appendix Fig. 3).

1. *Aloe barbadensis*, commonly known as Aloe vera, has been employed for its medicinal properties in the treatment of conditions like arthritis, radiation burns, diabetes, and ulcers (Eshghi et al. 2010; Guha et al. 2014; Hashemi et al. 2015). Aloe vera possesses anti-bacterial, anti-diabetic, and anti-inflammatory activities (Attah et al. 2016). This plant is rich in various vitamins, enzymes, and naturally active ingredients, including glycosides, acemannan,

anthraquinones, saponins, and oleic acid (Salehi et al. 2018). Notably, the enzymes in aloe vera contribute to pain and inflammation reduction, while salicylic acid, a key compound, exhibits pain-relieving properties akin to aspirin (Ahlawat and Khatkar n.d.; Gupta and Malhotra 2012; Hajhashemi et al. 2012).

2. *Salix alba*, or willow, has a historical reputation for pain relief. The bark extract of this plant contains salicin, which inhibits COX-2 activity and reduces the synthesis of pro-inflammatory mediators (Bonaterra et al. 2010; Drummond et al. 2013; Khayyal et al. 2005; Nahrstedt et al. 2007; Shara and Stohs 2015). Clinical evidence from a randomized, double-blind study suggests

that salicin, isolated from willow bark extract, can effectively alleviate low back pain (Chrubasik et al. 2000).

3. *Zingiber officinale*, commonly known as ginger, is a rich source of bioactive compounds like beta-carotene, curcumin, capsaicin, gingerols, caffeic acid, and salicylate (Altman and Marcussen 2001). Traditionally, ginger has found use in the treatment of various ailments, including rheumatism, gastrointestinal issues, and pain (Lantz et al. 2007). Research has shown that daily consumption of zingiber officinale can reduce pain and muscle injury induced by exercise (Black et al. 2010). Furthermore, a randomized controlled trial demonstrated the efficacy of ginger essential oil in reducing the intensity of low back pain when used in Swedish massage (Sritoomma et al. 2014).
4. *Curcuma longa*, commonly known as turmeric, has a long history of use as a spice, coloring agent, and flavoring agent. Its anti-inflammatory properties have made it a traditional remedy for conditions like arthritis (Al-Reza et al. 2010) curcumin, a key ingredient in curcuma longa, inhibits the synthesis of enzymes like LOX and COX, as well as various inflammatory cytokines and joint-destructive enzymes (Ramadan et al. 2010). Additionally, curcumin exhibits antioxidant properties by reducing plasma malondialdehyde levels and enhancing the activity of catalase and superoxide dismutase (Yuliani et al. 2018). Clinical studies have shown that daily intake of curcumin, in combination with other natural substances, can effectively reduce chronic low back pain without adverse effects (Qu et al. 2017).

Neuropathic pain

Neuropathic pain is a chronic condition characterized by sensory abnormalities, including hyperalgesia, dysesthesia, and allodynia (Amin et al. 2015). It affects a significant portion of the population, with a prevalence of 13–17%, particularly among women aged 50–64 and individuals in rural areas (Van Hecke et al. 2014). Factors contributing to neuropathic pain range from metabolic disorders like peripheral diabetic neuropathy to viral infections and inflammatory disorders (Colloca et al. 2017).

The etiology of neuropathic pain involves nociceptor sensitization, with alterations leading to neuroplasticity and central sensitization (Julius and Basbaum 2001). Receptors such as N-methyl-D-aspartate (NMDA) and glutamic metabotropic receptors play crucial roles in intracellular changes (Attal 2012). Various signaling pathways, including cAMP, protein kinases, and nitric oxide, are implicated in functional and structural alterations (Hucho and Levine 2007). Lipid metabolites like lysophosphatidic acid (LPA) released after tissue injury contribute to neuropathic pain through G-protein coupled LPA receptors (Ueda 2008). Inflammatory mediators such as interleukin, TNF-alpha, and nerve growth factor (NGF) further intensify hyperalgesia and allodynia (Sommer and Kress 2004). Treatment options for neuropathic pain encompass a range of drugs, including antidepressants, NSAIDs, opioids, serotonin and non-adrenaline reuptake inhibitors (SNRIs), anti-COX2 inhibitors, and gabapentin (Finnerup et al. 2015). Tricyclic antidepressants (TCAs) like amitriptyline are particularly effective due to their ability to block NMDA receptors, interfering with central sensitization (Kremer et al. 2016).

However, long-term use of certain medications can lead to adverse effects, such as weight gain, dizziness, and somnolence with TCAs and SNRIs, local irritation and pain with lidocaine and capsaicin, and cardiovascular and gastrointestinal issues with NSAIDs (Carter et al. 2014; Finnerup et al. 2010). Opioids may cause constipation, addictive behavior, and tolerance (Garland 2014).

Alternative approaches involve the use of medicinal plants and secondary metabolites with antioxidant and anti-inflammatory properties. These substances modulate protein kinases and CNS neurotransmitter systems, contributing to neuropathic pain management (Alonso-Castro et al. 2017; Juárez-Vázquez et al. 2013). Various medicinal plants and their active phytoconstituents have shown promise in alleviating neuropathic pain (Moreno-Salazar et al. 2008; Robles-Zepeda et al. 2011; Sanadgol et al. 2017).

In summary, neuropathic pain is a complex condition with diverse etiological factors and treatment options. While conventional medications can be effective, their long-term use may lead to adverse effects. Exploring alternative therapies, including medicinal plants and their active compounds, presents a promising avenue for pain management.

Table 4 Plant based interventions for Neuropathic pain

Plant (scientific name)	Bioactive compounds	Mechanism of action	Extract	Dose	Experimental model	results
1. <i>Acorus calamus</i>	Beta asarone, eugenol (Balakumbahan et al. 2010; Raja et al. 2009)	Improvement of neuron organelles and synaptic structure (Li et al. 2017a, b)	Saponin rich extract (Muthuraman and Singh 2012)	(200 and 40)mg/kg (Muthuraman and Singh 2012)	Rat (Muthuraman and Singh 2012)	Improves CCI induced sciatic functional changes, electrophysiological changes and nociceptive threshold levels (Muthuraman and Singh 2012)
2. <i>Alstonia scholaris</i>	Kaempferol (Khyade and Vaikos 2011)	Inhibits ROS and inflammatory cytokines production (H. Singh et al. 2017)	Methanolic extract (Singh et al. 2017)	10 mg/kg (Singh et al. 2017)	Rat (Singh et al. 2017)	Reduces the painful condition of heat and mechanical hyperalgesia and cold allodynia (Singh et al. 2017)
3. <i>Butea monosperma</i>	Butrin, isobutrin, coreopsin (Kasture et al. 2002; Surin and Ananthaswamy 2011)	Exhibits anti-oxidant, neuroprotective action and modulates and cellular calcium (Thiagarajan et al. 2012a)	Ethanollic extract (Thiagarajan et al. 2012a)	(200, 300 and 400) mg/kg (Thiagarajan et al. 2012a)	Rat (Thiagarajan et al. 2012a)	Decreases CCI induced histopathologica, biochemical and behavioral changes (Thiagarajan et al. 2012a)
4. <i>Curcuma longa</i>	Curcumin (X. Zhao et al. 2012)	Decreases serum COX2 concentrations (Zanjani et al. 2014)	Curcumin suspended in ethylolate (Zanjani et al. 2014)	(50 mg/kg (Zanjani et al. 2014)	Rat (Zanjani et al. 2014)	decreased mechanical and cold allodynia (Zanjani et al. 2014)
5. <i>Salvia divinorum</i>	Salvinorin A (McCurdy et al. 2006)	Activates kappa-opioid receptors (McCurdy et al. 2006)	Acetonic extract (Simón-Arceo et al. 2017)	(30, 100 and 200) mg/kg (Simón-Arceo et al. 2017)	Rat (Simón-Arceo et al. 2017)	Activates kappa-opioid receptors (Simón-Arceo et al. 2017)
6. <i>Crocus sativus</i>	Crocin (Alavizadeh and Hosseinzadeh 2014)	Decreases Glutamate, Aspartate in extra cellular space	Ethanollic and aqueous extract (Amin and Hosseinzadeh 2015)	200 mg/kg (Amin and Hosseinzadeh 2015)	Rat (Amin and Hosseinzadeh 2015)	Attenuates CCI induced hyperalgesia and allodynia (Amin and Hosseinzadeh 2015)
7. <i>Nigella sativa</i>	Thymoquinone (Forouzanfar et al. 2014; Kanter 2007)	Improves behavioral signs and oxidative effect of neuropathic pain (Amin et al. 2014)	Ethanollic extract (Tewari et al. 2015)	(500 and 1000) mg/kg (Tewari et al. 2015)	Rat (Tewari et al. 2015)	Exhibit analgesic effect on cisplatin induced neuropathic pain (Tewari et al. 2015)
8. <i>Ocimum sanctum</i>	Saponin (Kaur et al. 2015)	Attenuates the amount of nerve injury inciting agent induced calcium and free radicals (KAUR et al. 2015)	Methanolic leave extract (Muthuraman et al. 2008)	(100 and 200) mg/kg (Muthuraman et al. 2008)	Rat (Muthuraman et al. 2008)	Reduces thermal and mechanical hypernociception and allodynia (Muthuraman et al. 2008)
9. <i>Momordica charantia</i>	Gallic acid (Grover and Yadav 2004)	Reduces total calcium, GSH and superoxide ion (Kaur and Muthuraman 2019)	Fruit extract (Jain et al. 2015)	(400 and 800) mg/kg (Jain et al. 2015)	Rat (Jain et al. 2015)	Improves vincristine induced neuropathic pain (Jain et al. 2015)
10. <i>Rosmarinus officinalis</i>	Ursolic acid, caffeic acid, rosmarinic acid, carnosic acid (Yu et al. 2013)	Reduced activation markers (Iba1, GFAP), inflammatory factors - (TNF- α , iNOS, toll-like receptor 4), apoptosis-related mediators (Bax, cleaved caspase-3, and 9) (Ghasemzadeh et al. 2016)	Alcoholic extract (Ghasemzadeh et al. 2016)	(100, 200 and 400) mg/kg (Ghasemzadeh et al. 2016)	Rat (Ghasemzadeh et al. 2016)	Reduces hyperalgesia,, mechanical and cold allodynia (Ghasemzadeh et al. 2016)
10 <i>Rubia cordifolia</i>	Napthaquinone, anthraquinone (Devi Priya and Siril 2014)	GABA or Antioxidant inhibition	Alcoholic root and rhizome extract (Diwane et al. 2015)	(100, 200 and 400) mg/kg (Diwane et al. 2015)	Rat (Diwane et al. 2015)	Decreases the withdrawal latency in cold allodynia (Diwane et al. 2015)
11. <i>Salvia officinalis</i>	Carnosol and ursolic acid (Rodrigues et al. 2012)	Shows modulatory effects on TRPA-1 receptors (Rodrigues et al. 2012)	Butanol and aqueous extract (Qnais et al. 2010)	(10, 31.6, 100, 316, 1000) mg/kg (Qnais et al. 2010)	Rat (Qnais et al. 2010)	Increases the latency on hot plate assay (Qnais et al. 2010)

Table 4 continued

Plant (scientific name)	Bioactive compounds	Mechanism of action	Extract	Dose	Experimental model	results
12. <i>Choisya ternata</i>	Isopropyl, propyl-N-methyl anthranilates (Pinheiro et al. 2014)	Activates adrenergic and nitergic pathways, K ⁺ ATP channels and serotonergic pathways (Pinheiro et al. 2014)	Essential oil extracts (Pinheiro et al. 2015)	(3,10 and 30) mg/kg (Pinheiro et al. 2015)	Mice (Pinheiro et al. 2015)	Reduces NO, IL-1 beta and TNF alpha production (Pinheiro et al. 2015)
13. <i>Heterotheca inuloides</i>	7-hydroxy-3,4 dihydrocadalin (Rodríguez-Chávez et al. 2015)	Activates 5HT _{1A} , 5HT _{1D} receptors (Rocha-González et al. 2010)	Hexane extract (Rocha-González et al. 2010)	1–1000 µg/paw	Rat (Rocha-González et al. 2010)	Inhibited inflammation and arthritis (Rocha-González et al. 2010)
14. <i>Artemisia dracuncul</i>	Eugenol (Obolskiy et al. 2011)	Improvement of neuron organelles and synaptic structure (Li et al. 2017a, b)	Ethanolic extract (Obrosova et al. 2010)	500 mg/kg/day (Obrosova et al. 2010)	Rat (Obrosova et al. 2010)	Upregulates 12/15 LOX and expression of nitrated protein in peripheral nervous system (Obrosova et al. 2010)
15. <i>Pterodon pubescens</i>	Geranyl geraniol (Russo and Cardoso 2015)	Acts on 5HT ₃ receptors (Spindola et al. 2011)	Ethanolic fruit extract (Nucci-Martins et al. 2015)	(30–300) mg/kg (Nucci-Martins et al. 2015)	Mice (Nucci-Martins et al. 2015)	Prevents PSNL induced thermal and mechanical hyperalgesia (Nucci-Martins et al. 2015)
16. <i>Cannabis sativa</i>	Cannabidiol (Russo and Guy 2006)	Reduces NO, PGE2, GSH related enzymes and lipid peroxides (Costa et al. 2007)	<i>Cannabis sativa</i> extract (Comelli et al. 2008)	cannabis extract d 64.5% CBD (Comelli et al. 2008)	Rat (Comelli et al. 2008)	Attenuates vanilloid and TRPV1 induced thermal hyperalgesia (Comelli et al. 2008)
17. <i>Swietenia humilis</i>	Mexicanolide compound (Ovalle-Magallanes et al. 2017)	Activates GABaregic, NO, opiodergic and serotonergic pathways, guanylyl cyclase, K _{ATP} channels and inhibits hyperalgesia (Ovalle-Magallanes et al. 2017)	Aqueous extract (Ovalle-Magallanes et al. 2017)	(10, 31.6, 56.2, 100 and 177) microgram/paw (Ovalle-Magallanes et al. 2017)	Mice (Ovalle-Magallanes et al. 2017)	Activates GABaregic, NO, opiodergic and serotonergic pathways, guanylyl cyclase, K _{ATP} channels and inhibits hyperalgesia (Ovalle-Magallanes et al. 2017)

Various medicinal plants and their active phyto-constituents have shown promise in alleviating pain and addressing related symptoms. These natural remedies offer potential alternatives to conventional medications, which often come with adverse effects (Table 4, Fig. 2, online Appendix Fig. 4).

1. *Acorus calamus*, a member of the Araceae family, has been traditionally used in Indian medicine to manage severe inflammatory disorders. It contains phytoconstituents like beta asarone and eugenol, which have demonstrated benefits in reducing neuropathic pain and associated oxidative stress (Balakumbahan et al. 2010; Mittal et al. 2009). Additionally, hydroalcoholic extracts of *Acorus calamus* have been found to mitigate thermal hyperalgesia and allodynia in neuropathic pain models (Muthuraman et al. 2010).
2. *Alstonia scholaris*, known as the ‘Indian devil tree,’ has shown anti-nociceptive and anti-inflammatory properties due to its bioactive constituents, including coumarin, alkaloids, and flavonoids (Khyade and Vaikos 2011). Methanolic extracts of this plant have demonstrated effectiveness in reducing neuropathic pain symptoms and oxidative stress (Singh et al. 2017).
3. *Butea monosperma*, also known as the ‘Flame of the forest,’ *Butea monosperma* has exhibited neuroprotective effects, particularly in vincristine-induced neuropathic pain. Its bioactive compounds, such as butrin and isobutrin, contribute to the reduction of hyperalgesia and allodynia, possibly through their antioxidant properties (Thiagarajan et al. 2012b).

4. *Curcuma longa* with its active phenolic ingredient Curcumin, has been effective in alleviating neuropathic pain, including allodynia and hyperalgesia, while reducing COX-2 levels (Zanjani et al. 2014). This plant has shown potential as a natural remedy for neuropathic pain management.
5. *Salvia divinorum*, a native Mexican plant, has demonstrated efficacy in relieving neuropathic pain. Salvinorin A, one of its phytoconstituents, activates kappa-opioid receptors, providing analgesic effects in various animal models (McCurdy et al. 2006). Ethyl acetate extracts from *Salvia divinorum* have also shown antinociceptive properties through the activation of opioid and 5-HT_{1A} receptors (Tlacomulco-Flores et al. 2020).
6. *Crocus sativus*, *Crocus sativus*, commonly known as saffron, contains bioactive constituents like crocin and safranal, which have demonstrated potential in reducing neuropathic pain behavioral symptoms (Amin and Hosseinzadeh 2012). These extracts have also exhibited anti-inflammatory effects by reducing pro-inflammatory cytokines (Safakhah et al. 2016).
7. *Nigella sativa*, known for its volatile oil containing thymoquinone, has shown promise in improving neuropathic pain symptoms and oxidative effects (Amin et al. 2014). Ethanolic extracts of *Nigella sativa* have also demonstrated analgesic effects in cisplatin-induced neuropathic pain (Tavakkoli et al. 2017).
8. *Ocimum sanctum*, widely distributed in India, has been found to reduce neuropathic pain by decreasing calcium levels and oxidative stress induced by nerve injury (Kaur et al. 2015). Saponin, isolated from *Ocimum sanctum*, also plays a role in reducing painful neuropathic conditions (Kaur et al. 2015).
9. *Momordica charantia*, known as bitter melon, has shown neuroprotective actions, reducing hyperalgesia and allodynia, possibly through its anti-inflammatory and antioxidative activity (Jain et al. 2014).
10. *Rosmarinus officinalis* also known commonly as Rosemary, rich in phytoconstituents like rosmarinic acid, has demonstrated anti-inflammatory effects by reducing COX-2 and pro-inflammatory cytokines in neuropathic pain models (Ghasemzadeh Rahbardar et al. 2017).
11. *Rubia cordifolia*, an ayurvedic herb, exhibits anti-inflammatory and analgesic activities, possibly through GABAergic or antioxidant mechanisms (Diwane et al. 2015).
12. *Salvia officinalis*, garden sage, containing carnosol and ursolic acid, has strong antinociceptive effects and may modulate TRPA-1 receptors (Rodrigues et al. 2012).
13. *Choisya ternata*, has antinociceptive effects due to its active constituents, such as isopropyl, methyl, and propyl-N-methyl anthranilate, which activate various pathways (Pinheiro et al. 2014).
14. *Heterotheca inuloides*, also known as ‘Arnica Mexicana,’ has shown anti-neuropathic action by activating opioid and serotonergic receptors, as well as guanylyl cyclase (Rocha-González et al. 2014).
15. *Artemisia dracuncululus*, exhibits anti-nociceptive and anti-inflammatory activity, possibly through eugenol and other phytoconstituents (Abad et al. 2011).
16. *Pterodon pubescens*, prevents thermal and mechanical hyperalgesia by acting on 5HT₃ receptors (Spindola et al. 2011).
17. *Cannabis sativa*, commonly known as marijuana, contains cannabidiol, which alleviates neuropathic pain and reduces oxidative stress (Costa et al. 2007; Serpell et al. 2014).
18. *Swietenia humilis*, grown in Mexico, inhibits hyperalgesia by activating GABAergic, NO, opioidergic, and serotonergic pathways, as well as guanylyl cyclase and KATP channels (Ovalle-Magallanes et al. 2017).

Dysmenorroehea

Dysmenorrhea is a common menstrual disorder that affects approximately 50% of women of reproductive age (Bernardi et al. 2017). The pain experienced during dysmenorrhea can range from mild to severe and is often accompanied by symptoms such as diarrhea, nausea, vomiting, and constipation (Edmonds 2012). Dysmenorrhea can significantly impact a woman's quality of life and social activities (Mrugacz et al. 2013).

In the treatment of dysmenorrhea, inhibitors of prostaglandin synthesis and NSAIDs like aspirin, ibuprofen, and mefenamic acid are commonly used due to their ability to inhibit thromboxane and prostaglandins (García-Martínez et al. 2015). However, it's important to note that these drugs can have adverse effects, including hepatotoxicity and bone loss (Williams et al. 2013) and gastrointestinal and liver issues (Moll et al. 2011).

In addition to conventional medications, many medicinal plants have been explored as alternative treatment strategies for dysmenorrhea due to their anti-spasmodic, analgesic, and prostaglandin inhibitory effects. These natural remedies offer potential options for managing this common and often debilitating condition. Further research and critical appraisal of the literature are essential to assess the effectiveness and safety of these alternative treatments in the management of dysmenorrhea.

Various medicinal plants have been explored for their potential in alleviating dysmenorrhea, a common and often debilitating condition in women. These plants offer promising anti-spasmodic, analgesic, and prostaglandin inhibitory effects. Here, we summarize the key findings from the literature on selected plants, their active compounds, mechanisms of action, and recommended doses (Table 5, Fig. 2, online Appendix Fig. 5).

1. *Zingiber officinalis* (Ginger), containing compounds like gingerol and gingerdiol, has been shown to inhibit the synthesis of prostaglandins and inflammatory cytokines. It effectively relieves pain, including primary dysmenorrhea. A dose of 1500 mg/day of *Zingiber officinalis* root powder has been found to effectively alleviate severe pain associated with primary dysmenorrhea (Rahnama et al. 2012). Additionally, a mixture of 750 mg/day *Zingiber officinalis* powder and zinc sulfate showed significant pain relief compared to a placebo (Kashefi et al. 2014).
2. *Foeniculum vulgare* (Fennel, with active compounds like estragole and trans-anethole, reduces uterine contractions by inhibiting prostaglandins. A 2% fennel oil drop has shown effectiveness in reducing primary dysmenorrhea pain when compared to mefenamic acid (Bokaie et al. 2013). Combining *Foeniculum vulgare* with vitex has been found to be more effective in reducing dysmenorrhea symptoms than mefenamic acid (Zeraati et al. 2014).
3. *Cinnamomum zeylanicum* (Cinnamon), contains eugenol, which inhibits prostaglandin synthesis and inflammation. It has been shown to reduce dysmenorrhea pain as effectively as ibuprofen when taken at a dose of 1260 mg/day (Jaafarpour, Hatefi, Khani, et al., 2015). Cinnamon also reduces nausea, menstrual bleeding, and vomiting in primary dysmenorrhea without adverse effects (Jaafarpour et al. 2015a, b).
4. *Trigonella foenum-graecum* (Fenugreek), contains gamma-linolenic acid, which inhibits prostaglandin synthesis and cyclooxygenase pathway, reducing pain. A dose of 6 g/day of *Trigonella foenum-graecum* has been found to be as effective as mefenamic acid in reducing the severity of dysmenorrhea (Inanmdar et al. 2016). Capsules prepared from fenugreek seed powder at a dose of 2700 mg/day have also shown efficacy in relieving dysmenorrhea symptoms (Younesy et al. 2014).
5. *Mentha piperita* (Peppermint), contains menthol, which inhibits prostaglandin synthesis and myometrium contractions. Peppermint oil has been shown to be safer and more effective than mefenamic acid in relieving dysmenorrhea pain (Masoumi et al. 2016). A dose of 990 mg/day of *Mentha piperita* capsules has been effective in reducing the pain of primary dysmenorrhea (Heshmati et al. 2016).
6. *Teucrium polium*, has anti-inflammatory and antispasmodic properties. The aqueous extract of *Teucrium polium* at a dose of 1000 mg/day during menstruation reduced the pain and severity of primary dysmenorrhea as effectively as mefenamic acid (Abadian et al. 2016).
7. *Triticum aestivum* (Wheat Germ), contains vitamins that attenuate pain in primary dysmenorrhea patients. An extract dose of 1200 mg/day taken during the menstrual cycle's specified days has been shown to alleviate the symptoms of primary dysmenorrhea (Ataollahi et al. 2014).
8. *Rosa damascene* (Rose), with its high vitamin C and flavonoid content, has analgesic and anti-inflammatory effects. A dose of 600–800 mg/day of *Rosa damascene* has been found to be as effective as mefenamic acid in reducing

Table 5 Plant based interventions for Dysmenorrhoea

Plant (Scientific name)	Bioactive compounds	Mechanism of actions	Extract	Dose	Experimental model	Results
1. <i>Zingiber officinalis</i>	Gingerdiol, gingerol, caffeic acid (Semwal et al. 2015), curcumin (Daily et al. 2015)	Inhibit prostaglandin, inflammatory cytokines, leukotriene synthesis and thromboxane activity (Chen et al. 2016; Mirabi et al. 2014)	Powder of <i>Zingiber</i> root (Rahnama et al. 2012)	1500 mg/kg (Rahnama et al. 2012)	Human (Rahnama et al. 2012)	Relieves severe pain of primary dysmenorrhoea (Rahnama et al. 2012)
2. <i>Foeniculum vulgare</i>	Essential oil (Badgujar et al. 2014; Mirabi et al. 2014; Rather et al. 2016)	Inhibit prostaglandin and circulatory oxytocin levels and reduces uterine contraction (Badgujar et al. 2014; Mirabi et al. 2014; Rather et al. 2016)	Capsules of <i>Foeniculum</i> (Moslemi et al. 2012)	184 mg/day (Moslemi et al. 2012)	Human (Moslemi et al. 2012)	Reduces pain intensity of primary dysmenorrhea (Moslemi et al. 2012)
3. <i>Cinnamomum zeylanicum</i>	Eugenol (Mirabi et al. 2014; Nabavi et al. 2015)	Inhibits prostaglandin synthesis and inflammation (De Smet et al. 1993)	capsule containing 420 mg Cinnamon (Jaafarpour et al. 2015)	1260 mg/day (Jaafarpour et al. 2015)	Human (Jaafarpour et al. 2015)	Reduced severity score and duration of pain (Jaafarpour et al. 2015)
4. <i>Trigonella foenum-graecum</i>	Gamma linolenic acid (Inanmdar et al. 2016)	Inhibits prostaglandin synthesis and cyclooxygenase pathways (Inanmdar et al. 2016)	Capsules of seed powder (Younesy et al. 2014)	2700 mg/day (Younesy et al. 2014)	Human (Younesy et al. 2014)	Relieves the severity and symptoms of primary dysmenorrhoea (Younesy et al. 2014)
5. <i>Mentha piperita</i>	Menthol, menthyl acetate, cineole, menthofurane (Heshmati et al. 2016; Kamatou et al. 2013)	Inhibits prostaglandin, leukotriene and interleukine synthesis (Heshmati et al. 2016; Kamatou et al. 2013)	Peppermint capsule (Heshmati et al. 2016)	990 mg/day (Heshmati et al. 2016)	Human (Heshmati et al. 2016)	Alleviates the severity of dysmenorrhea (Heshmati et al. 2016)
6. <i>Teucrium polium</i>	Alpha-pinene, ocemene, pulegone (Abadian et al. 2016)	inhibiting acetylcholine-induced muscle contraction (Bahramikia & Yazdanparast 2012)	<i>Teucrium polium</i> capsule (Abadian et al. 2016)	1000 mg/day (Abadian et al. 2016)	Human (Abadian et al. 2016)	Decreases the pain of primary dysmenorrhea (Abadian et al. 2016)
7. <i>Triticum aestivum</i>	Alpha tocopherol (Güven & KARA, 2016)	uses pro -inflammatory activities to activate neuropeptides, cytokines, and macrophages and reduces inflammation (Ataollahi et al. 2014)	Capsules of germ extract (Ataollahi et al. 2014)	1200 mg/day (Ataollahi et al. 2014)	Human (Ataollahi et al. 2014)	Alleviates the symptoms of primary dysmenorrhoea (Ataollahi et al. 2014)
8. <i>Rosa damascene</i>	Geraniol, citranello, kaempferol (Bani et al. 2014)	Regulating acetylcholine release or Stimulating GABAergic and putative neurons	<i>Rosa damascene</i> capsule (Bani et al. 2014)	800 mg/day (Bani et al. 2014)	Human (Bani et al. 2014)	Reduces the intensity of primary dysmenorrhea (Bani et al. 2014)
9. <i>Melissa officinalis</i>	Citral, alpha-pinene, rosmarinic acid (Miraj et al. 2016)	Through cholinergic system and L-Arginine-nitric pathway	<i>Melissa officinalis</i> capsule (Mirabi et al. 2018)	990 mg/day (Mirabi et al. 2018)	Human (Mirabi et al. 2018)	Decreases the severity of dysmenorrhea (Mirabi et al. 2018)
10. <i>Thymus vulgaris</i>	Thymol, eugenol (Assiri et al. 2016)	Inhibits prostaglandin synthesis and inflammation (De Smet et al. 1993)	Essential oil (Salmalian et al. 2014)	25 drops (Salmalian et al. 2014)	Human (Salmalian et al. 2014)	Reduces pain severity (Salmalian et al. 2014)
11. <i>Valeriana officinalis</i>	Valtrate (Bone & Mills 2013)	Inhibits calcium ion binding to smooth muscle (Mirabi et al. 2011, 2014)	Powder of valerian root (Mirabi et al. 2011)	765 mg/day (Mirabi et al. 2011)	Human (Mirabi et al. 2011)	Decreases primary dysmenorrhea intensity (Mirabi et al. 2011)

Table 5 continued

Plant (Scientific name)	Bioactive compounds	Mechanism of actions	Extract	Dose	Experimental model	Results
12. <i>Rheum emodi</i>	Emodin, oxanthrone (Rehman et al. 2015; Takeoka et al. 2013)	Arachidonic acid release and COX2 enzyme expression	Capsules of <i>Rheum emodi</i> powder (Rehman et al. 2015)	840 mg/day (Rehman et al. 2015)	Human (Rehman et al. 2015)	Capable of reducing pain of dysmenorrhea (Rehman et al. 2015)
13. <i>Achillea millefolium</i>	Cosmossin 1, cyanaroside 1 (Saeidnia et al. 2011)	Flavonoids (spasmolytic agent) relaxes smooth muscle of mammals	<i>A.millefolium</i> powder (Jenabi and Fereidoony 2015)	4 g powder/ 300 ml water (Jenabi and Fereidoony 2015) 3 days/month	Human (Jenabi and Fereidoony 2015)	Minimises pain in primary dysmenorrhoea
14. <i>Anethum graveolens</i>	Sabinene and limonene (Jana and Shekhawat 2010; Kazemi 2015b)	Inhibits NO synthesis (Kazemi 2015a)	Capsules of seed powder (Heidarifar et al. 2014)	2000 mg/day (Heidarifar et al. 2014)	Human (Heidarifar et al. 2014)	Reduces pain intensity of primary dysmenorrhea (Heidarifar et al. 2014)
15. <i>Matricaria chamomilla</i>	Azoline, matrisin, bisabolol, apigenin, methoxy-coumarin (Gosztola et al. 2010; Orav et al. 2010; Srivastava et al. 2010)	Reduces prostaglandins (Radfar et al. 2018)	<i>Matricaria</i> capsule (Radfar et al. 2018)	750 mg/day (Radfar et al. 2018)	Human (Radfar et al. 2018)	Reduces pain intensity of dysmenorrhea (Radfar et al. 2018)

pain sensitivity during primary dysmenorrhea (Bani et al. 2014).

9. *Melissa officinalis* (Lemon Balm), possesses antispasmodic, anti-inflammatory, and analgesic effects. Capsules containing 990 mg/day of *Melissa officinalis* have been shown to decrease the severity of dysmenorrhea (Mirabi et al. 2018). The use of *Melissa officinalis* tea bags every 8 h after menstruation has been effective in alleviating dysmenorrheal symptoms (Safdari Dehcheshmeh and Parvin 2016).
10. *Thymus vulgaris* (Thyme), contains thymol and other compounds with antispasmodic and anti-inflammatory properties. Essential oil from *Thymus vulgaris*, taken at a specified dose, significantly improved primary dysmenorrhea pain compared to ibuprofen (Salmalian et al. 2014).
11. *Valeriana officinalis* (Valerian), has sedative and antispasmodic effects. A dose of 765 mg/day of valerian root has been shown to be more effective in treating primary dysmenorrhea than a placebo (Mirabi et al. 2011).
12. *Rheum emodi*, has analgesic and anti-inflammatory effects. Capsules prepared from the plant powder, taken at a dose of 840 mg/day during specified days of the menstrual cycle, have been

found to be as effective as mefenamic acid in reducing dysmenorrhea pain (Rehman et al. 2015).

13. *Achillea millefolium* (Yarrow), has antispasmodic and anti-prostaglandin effects. Tea made from *Achillea millefolium* has been effective in improving primary dysmenorrhea symptoms (Jenabi and Fereidoony 2015).
14. *Anethum graveolens* (Dill), has been shown to relieve the pain and severity of primary dysmenorrhea as effectively as mefenamic acid when taken at a dose of 2000 mg/day (Heidarifar et al. 2014).
15. *Matricaria chamomilla* (Chamomile), contains active compounds like bisabolol and apigenin, which have anti-inflammatory and antispasmodic effects. A dose of 250 mg of *Matricaria chamomilla* taken three times daily has been shown to reduce primary dysmenorrhea pain (Radfar et al. 2018).

Headache

Headache is a prevalent neurological problem that affects a substantial portion of the global population, making it one of the most common reasons for

Table 6 Plant based interventions for Headache

Plant (Scientific name)	Active compounds	Mechanism of actions	Extract	Dose	Experimental model	Result
1. <i>Sambucus nigra</i>	Citric acid, malic acid, shikimic acid (Dulf et al. 2013; Ma and Wu 2012), isoquercetin, P-coumaric acid, ferulic acid (Bhattacharya et al. 2013b)	Activation to hypothalamic-pituitary-axis. Releases adrenocortical hormone	Hydroalcoholic leave extract (Neekhara et al. 2021)	(200 and 400) mg/kg (Neekhara et al. 2021)	Male Wistar rats (Neekhara et al. 2021)	Reduces chronic stress induced perturbations (Neekhara et al. 2021)
2. <i>Allium cepa</i>	Quercetin (Sagar et al., 2020b)	Downregulates NF-KB levels, inhibits PGE2 and TXB2 levels derived from arachidonic acid (Moslemi et al. 2012)	Methanolic extract	(50, 250 and 500) microgram/ml	LPS-induced BV-2 microglial cells	Decreases the level of TNF-alpha, IL-6, IL-beta etc. proinflammatory cytokines
3. <i>Ligusticum chuanxiong</i>	Ligustilide (Zhao et al., 2014b; Zhu et al. 2014a, b)	Inhibits pro-inflammatory cytokines, NF-KB induced chemotaxis (L. X. Zhao et al., 2014b; Zhu et al. 2014a, b)	Ethanollic extract (Yuan et al. 2010)	32 mg/kg (Yuan et al. 2010)	Rat (Yuan et al. 2010)	Reduces headache by adjusting monoamine neurotransmitter (Yuan et al. 2010)
4. <i>Rosmarinus officinalis</i>	Rosmarinic acid, carnosol, betulinic acid, ursolic acid (Borrás-Linares et al., 2014b)	Inhibits uptake of monoamines and monoamine oxidase activity (Waggas and Balawi 2008)	Leaf extract (Waggas and Balawi 2008)	100 mg/kg (Waggas and Balawi 2008)	Rat (Waggas and Balawi 2008)	Show neuroprotective effects by decreasing dopamine, epinephrine, norepinephrine and catecholamine content (Waggas and Balawi 2008)

seeking medical treatment (Stovner et al. 2007). According to the third edition of the International Headache Society, headache is classified into primary and secondary. Cephalalgias, cluster headache, tension-type headaches (TTH), external pressure headache where cold stimulus headache, etc., are primary headaches and headaches caused by underlying diseases including vascular disorders or tumors, head injuries, and infections are secondary headaches (Headache Classification Committee of the International Headache Society 2013). Primary headaches, including TTH and migraine, are the most frequent in childhood (Genizi et al. 2013). Nonsteroidal anti-inflammatory drugs (NSAIDs) (Krishnasamy et al. 2020), including aspirin and ibuprofen (Hasnaoui et al. 2003), triptans (Kassem & Labib 2016), and ergot alkaloids (Krishnasamy et al. 2020), are used in the treatment of headaches, but none of these can fully recover the recurrent attacks (Table 6, Fig. 2, online Appendix Fig. 6).

1. *Sambucus nigra*: Elderberry, known as *Sambucus nigra*, is rich in vitamins, essential oils, and various minerals. It possesses antioxidant and anti-inflammatory properties. Elderberry tea is

traditionally used for pain relief, and studies have shown that its methanolic fruit extract can reduce headache severity (Ulbricht et al. 2014). The plant possesses anti-oxidant (Jimenez et al. 2014a; Topolská et al. 2015) and anti-inflammatory (Ulbricht et al. 2014) properties. The plant shows antidiabetic potential by promoting insulin secretion and glucose uptake (Bhattacharya et al. 2013a). The plant is effective in headache treatment.. The plant inhibits the synthesis of COX- and 10 µg/ml of its methanolic fruit extract reduces the severity of headache compared to an indomethacin control group (Thole et al. 2007).

2. *Allium cepa*: *Allium cepa*, commonly known as onion, is a plant of the Liliaceae family cultivated worldwide, particularly in countries with moderate climates (Bisen and Emerald 2016). *Allium cepa* possesses anti-tumor, antiarthritic, antimicrobial, anti-hyperlipidaemic, and anticarcinogenic properties (Upadhyay 2017). Active phytoingredients such as vitamins, quercetin, luteonin, kaempferol, etc., are present in onion (Sagar et al. 2020a) that downregulate NF-KB

levels, inhibit PGE2 and TXB2 levels derived from arachidonic acid (Lesjak et al. 2018), as well as chalcones, isoflavones, flavanones, and anthocyanins (Liguori et al. 2017).

3. *Ligusticum chuanxiong*: *Ligusticum chuanxiong* is a plant of the Umbelliferae family used in traditional Korean, Chinese and Japanese folk medicine systems (Kemper 2011). The plant has anti-inflammatory (Or et al. 2011), antioxidant (Ramalingam and Yong-Ki 2010) (J. B. Jeong et al. 2009) and neuroprotective effects (Shu et al. 2009) (Lin et al. 2009). Ligustilide isolated from this plant inhibits the production of pro-inflammatory cytokines, Nf-kB induced chemokines and thus helps in inflammatory pain reduction (Zhao et al., 2014a; Di Zhu et al. 2014a, 2014b).
4. *Rosmarinus officinalis*: *Rosmarinus officinalis* has been used to treat numerous diseases such as rheumatic pain, dysmenorrheal, epilepsy, stomachache, depression, and mental fatigue in folk medicine (Heinrich et al. 2006). The perennial shrub has anti-inflammatory (Benincá et al. 2011), neuroprotective (Hou et al. 2013), antioxidant (Bakirel et al. 2008) and memory-enhancing effects (Sasaki et al. 2013). The active ingredients of the plant include rosmarinic acid, carnosol, carnosic acid, betulinic acid, rosmanol and ursolic acid (Borrás-Linares et al. 2014a). Carnosol, betulinic acid, and ursolic acid isolated from *Rosmarinus officinalis* aerial crude extract inhibit leukocyte, IL-1beta, TNF-alpha, etc., pro-inflammatory mediators in carrageenan-induced murine pleurisy model (Benincá et al. 2011).

Migraine

Migraine, is characterized by recurrent headaches with or without aura, often beginning in childhood and peaking between the ages of 22 and 55 years, with a higher prevalence in women (Burststein et al. 2015). These debilitating headaches are associated with pulsating pain of moderate to severe intensity, often leading to impaired daily physical activity (Olesen 2018). Common accompanying symptoms include nausea, vomiting, sensitivity to light and sound, all localized to one-half of the head (Charles

and Hansen 2015). The headache itself is typically unilateral, throbbing, and can last for hours to days (Olesen 2018).

The etiology of migraine is multifactorial, involving lifestyle, environmental, and genetic factors (Durham and Papapetropoulos 2013). External triggers include alcoholic and caffeinated beverages (Millichap and Yee 2003), smoking and stress (Martin 2010). Additionally, sex hormones, particularly estradiol and progesterone, play a role in menstrual migraines (Gupta et al. 2007). Treatment options for migraines encompass a range of pharmaceuticals. Aspirin, acetaminophen, ergot derivatives (Monteith and Goadsby 2010), beta-blockers like propranolol and metoprolol, and anticonvulsants such as valproate and topiramate are commonly employed (Goadsby and Sprenger 2010).

There is growing interest in alternative treatments for migraines, with numerous medicinal plants showing promise due to their safe and effective phytoingredients (Table 7, Fig. 2, online Appendix Fig. 7).

1. *Tanacetum parthenium* (Feverfew): Feverfew, widely distributed in North America, Europe, and South America, has been traditionally used for various ailments, including nausea and allergies. It contains parthenolide, a sesquiterpene lactone, which has anti-inflammatory and anti-platelet aggregation properties. Parthenolide inhibits histamine release, reduces nitroglycerin-induced Fos expression, and targets the TRPA1 channel, all contributing to its potential to prevent migraines (Pareek et al. 2011; Rios and Passe 2004; Sun-Edelstein and Mauskop 2011).
2. *Petasites hybridus* (Butterbur): Butterbur, found in Europe and Asia, has been traditionally used for various ailments, including asthma and fever. Its active compounds, petasin and isopetasin, inhibit leukotriene synthesis and COX2-mediated prostaglandin E2 release, offering relief from migraines. Additionally, isopetasin activates the TRPA1 channel, providing antinociceptive effects (Lipton et al. 2004; Sun-Edelstein and Mauskop 2011; Taylor 2011).
3. *Hypericum perforatum* (St. John's Wort): Known for its effectiveness in treating mild to moderate depression, *Hypericum perforatum* also

Table 7 Plant based interventions for Migraine

Plant (scientific name)	Bioactive compounds	Mechanism of actions	Extract	Dose	Experimental model	Results
1. <i>Tanacetum parthenium</i>	Parthenolide (Rajapakse and Pringsheim 2016)	Inhibits histamine release (Pareek et al. 2011) and NO synthesis (Aviram et al. 2012)	Water extract (Recinella et al. 2020)	(10–100) microgram/ml (Recinella et al. 2020)	Mice (Recinella et al. 2020)	Reduces inflammatory pathways and inhibits hypothalamic dopamine release (Recinella et al. 2020)
2. <i>Petasites hybridus</i>	Petasin and isopetasin (Prieto 2014; Taylor 2011)	Inhibits leukotriene synthesis and release of COX2 mediated PGE2 (Taylor 2011)	Root extract (Lipton et al. 2004)	50 mg/day (Lipton et al. 2004)	Human (Lipton et al. 2004)	Reduces the number of monthly migraine attacks (Lipton et al. 2004)
3. <i>Hypericum perforatum</i>	Hypericin, hypericum (Wölflé et al. 2013)	Counteracts NO donor induced by blocking NF-KB, STAT1 and CREB etc. protein kinase C mediated pathway (Galeotti & Ghelardini 2013)	<i>Hypericum perforatum</i> dried extract (Galeotti & Ghelardini 2013)	t(5 mg/kg p. o) (Galeotti and Ghelardini 2013)	Mouse (Galeotti and Ghelardini 2013)	Counteracts pain hypersensitivity (Galeotti and Ghelardini 2013)
3. <i>Ginkgo biloba</i>	Ginkgolide B (Zhou et al. 2013)	Modulates brain glutamatergic transmission and antagonize PAF receptor (Tulsulkar and Shah 2013)	Leaf extract (Tulsulkar and Shah 2013)	100 mg/kg (Tulsulkar and Shah 2013)	Mice (Tulsulkar and Shah 2013)	Reduces hippocampal neuronal death (Tulsulkar and Shah 2013)
4. <i>Mentha piperita</i>	Menthol (Heshmati et al. 2016)	Inhibits prostaglandin, leukotriene and interleukin synthesis (Heshmati et al. 2016)	ethanol solution (Haghighi et al. 2010)	10% ethanol solution (Haghighi et al. 2010)	Human (Haghighi et al. 2010)	Alleviated symptoms of migraine without aura
5. <i>Zingiber officinalis</i>	6-Paradol (Gaire et al. 2015)	Attenuates neuroinflammation in microglia (Gaire et al. 2015)	Ginger rhizome powder (Gaire et al. 2015)	250 mg/day (Gaire et al. 2015)	Human (Gaire et al. 2015)	Reduces the common migraine attack (Gaire et al. 2015)
6. <i>Matricaria chamomilla</i>	Apigenin, bisabolol (Gosztola et al. 2010)	Inhibits iNOS expression, NO synthesis and PGE2 release (Zargaran et al. 2014)	Aqueous extract in sesame oil (Zargaran et al. 2014)		Human (Zargaran et al. 2014)	Pain relief in migraine patients (Zargaran et al. 2014)
7. <i>Cannabis sativa</i>	Cannabidiol (Izzo et al. 2009)	Inhibits cytokines, chemokines and ROS production (Zou and Kumar 2018)	Ethanol 96%	5 mg/day up to 12 days	Human (18–64) years old	Relief muscle stiffness, sensitivity, sleep quality, relief body pain and spasm

possesses analgesic properties. Its major active compounds, hypericin and hyperforin, exhibit anti-inflammatory effects and can counteract nitric oxide-induced pain hypersensitivity (Galeotti et al. 2010a, b; Kasper et al. 2010).

4. *Ginkgo biloba* leaf extracts contain terpene lactones and flavonoids, with ginkgolide B being particularly noteworthy. Ginkgolide B has antioxidant and anti-inflammatory effects, reducing neuronal damage and modulating platelet-activating factor (PAF) release. This modulation can be especially beneficial in migraine management (Ahlemeyer and Krieglstein 2003; Choi et al. 2019; D'Andrea et al. 2009; Tulsulkar & Shah 2013).
5. *Mentha piperita* (Peppermint) active compound, menthol, inhibits prostaglandin and interleukin synthesis, acting as a pain reliever. Menthol also activates TRPM8 receptors, providing relief from meningeal inflammation-induced pain (Haghighi et al. 2010; Heshmati et al. 2016; Jimenez et al. 2014b; Kamatou et al. 2013).
6. *Zingiber officinale* (Ginger) contains gingerols, diterpenoids, and flavonoids that inhibit the synthesis of prostaglandins and inflammatory cytokines. It also offers anti-inflammatory and antibacterial effects. 6-paradol, a compound found in ginger, attenuates neuroinflammation, making it a potential treatment option (Maghbooli et al. 2014; Mintah et al. 2019; Nour et al. 2017).
7. *Matricaria chamomilla* (Chamomile): Chamomile has been used traditionally for various purposes, including pain relief. It contains active compounds like bisabolol and apigenin, which inhibit iNOS expression and PGE2 release, providing anti-inflammatory effects. Chamomile gel has shown efficacy in reducing migraine symptoms (Miraj and Alesaeidi 2016; Zargaran et al. 2014, 2018).
8. *Cannabis sativa* (Cannabis), contains various phytocannabinoids like CBD and THC. CBD has anti-inflammatory properties, modulating pain perception and reducing cytokine release. It interacts with receptors like TRPV1 and 5-HT1A, providing relief from inflammatory and neuropathic pain. Additionally, CBD reduces

nitroglycerin-induced neuronal activation and central sensitization (Russo et al. 2008; Stevens et al. 2016; Zou & Kumar 2018).

Chest pain

The chest is an important part of our body and is located between the ribs from which various problems of our body can occur (Goodacre et al. 2005). Chest pain is one of the most frequent symptoms for which patients contact emergency medical services (EMS) and emergency departments (ED) (Mockel et al. 2013; Burman et al. 2011). Chest pain can be caused due to cardiac causes, chest wall diseases, gastrointestinal symptoms, pulmonary causes, mental and psychological causes, and non-cardiac chest pain (das Virgens et al. 2017; Webster et al. 2014). Few medicinal plants and their mechanism of action to prevent chest pain are discussed in Table 8, Fig. 2, online Appendix Fig. 8).

1. *Salvia miltiorrhiza*: *Salvia miltiorrhiza*, commonly known as red sage, is a perennial plant of the Lamiaceae family (Yuan Fang et al. 2008). Traditionally, this plant is used in the treatment of cardiovascular diseases (Luo et al. 2015) such as thrombosis (Wu & Wang, 2012a), atherosclerosis, and angina pectoris (Yang et al. 2006). The primary active compounds of this plant include salvianolic acid and diterpenoid tanshinones that are hydrophilic and lipophilic, respectively (Guo et al. 2014; Ho & Hong 2011; Zhao et al. 2008; Fan et al. 2009a, b). Other active compounds include Tanshinone II-A and cryptotanshinone (Tian et al. 2017a). By inhibiting the STAT-3 pathway and expressing MMP-9, cryptotanshinone reduces the heartbeat at a dose of 10 mg/kg/d for 28 days in diabetic rats (Tian et al. 2017a). In this way, the plant prevents cardiovascular disease-related chest pain.
2. *Carthamus tinctorius*: Commonly known as Safflower, *Carthamus tinctorius* is a member of the Asteraceae family (Sciences and 2010, n.d.). Flavonoids such as safflavinonesides A, safflavinonesides B (He et al. 2014a), alkaloids including guanosine (Fan et al. 2009a, b), saffloquinoside A and saffloquinoside B, and two

Table 8 Plant based interventions for Chest pain

Plant (Scientific name)	Bioactive compounds	Mechanism of actions	Extract	Dose	Experimental model	Results
1. <i>Salvia miltiorrhiza</i>	Tanshione II A and cryptotanshinone (Tian et al. 2017b)	Inhibits STAT 3 pathway and MMP-9 expression (Tian et al. 2017b)	cryptotanshinone	10 mg/kg/d cryptotanshinone	Rats	Reduces cardiac fibrosis
2. <i>Carthamus tinctorius</i>	Safflloflavonesides A and B, saffloquinoside A and B (Chen et al. 2013; He et al. 2014b)	Suppresses c-Jun N-terminal kinase and inhibits lipopolysaccharide (LPS)-induced apoptosis in cardiomyoblast cells (He et al. 2014b)	ethanolic extract (He et al. 2014b)	31.25, 62.5, and 125 µg/mL	H9c2 cardiomyoblast cells	Dilating coronary artery, Myocardial Ischemia improvement, Antithrombosis, Immune system modulating
3. <i>Allium sativum</i>	Alliin (Zeng et al. 2017b)	Inhibits plaque atherosclerosis and increases vasodilator (Ilmawati et al. 2017b)	Methanolic extract (Jeong et al. 2016b)	1000 µg/ml (Jeong et al. 2016b)	RAW 264.7 cells (Jeong et al. 2016b)	Prevents COX2 and PGE 2 production (Jeong et al. 2016b)
4. <i>Calendula officinalis</i>	Kaempferol, quercetin and isorhamnetin (Re et al., 2009b)	downregulation of pro-inflammatory cytokines like TNF-, IL-1, IL-6 (Ray et al. 2010)	<i>Calendula officinalis</i> in KHB buffer solution (Ray et al. 2010)	50 mM (Ray et al. 2010)	Rat (Ray et al. 2010)	Stimulates left ventricular pressure and aortic flow, reduces myocardial infraction size and apoptosis of cardiomyocytes (Ray et al. 2010)

quinochalcons (Jiang et al. 2010) have been isolated from this plant and have numerous biological activities. *Carthamus tinctorius* injection at a dose of 2.5 and 0.625 g/kg for 5 days inhibits the elevation of the typical ECGS-T segment, reduces serum IL-6 and TNF-alpha concentration, and suppresses the overexpression of Bax protein and Bcl-2 in rat (Han et al. 2009).

3. *Allium sativum*: *Allium sativum* or garlic is a plant of the Amaryllidaceae family. The major active phytoconstituents are allicin, E-ajoene, Z-ajoene (Sendl 1995), and alliin (Zeng et al. 2017a). By inactivating NF-kB, the methanolic extract of aged black garlic prevents the production of COX-2 and PGE-2 (Jeong et al. 2016a). Alliin inhibits plaque atherosclerosis and increases vasodilation (Ilmawati et al. 2017a).
4. *Calendula officinalis*: The major active ingredients of *calendula officinalis* are flavonoids such as kaempferol, quercetin, isorhamnetin, terpenoids, steroids, phenolic acids, carotenoids, etc. (Re et al. 2009a), which exhibit anti-inflammatory and antioxidant effects (Moghaddasi Mohammad and Kashanisup 2012). By activating protein kinase B and B-cell lymphoma 2 and decreasing TNF-alpha, *Callendula officinalis* changes the

death signal from ischemic reperfusion into a survival signal and thus protects the heart (Lecour et al. 2005).

Wounds

Wounds, a significant cause of disability are defined as physical injuries that disrupt the skin's normal functions or anatomy (Strodtbeck 2001). They often entail pain, infection risks, disfigurement, and unpleasant odors. Wounds are categorized as open or closed based on whether the skin remains intact (Lazarus et al. 1994; Percival 2002). Furthermore, wounds are classified into acute and chronic types depending on the wound healing process (Percival 2002). Acute wounds, like surgical incisions, typically heal through regular compensatory mechanisms (Percival 2002). In contrast, chronic wounds lead to pathologic inflammation and may require extended healing periods (Menke et al. 2007). Common causes of chronic wounds include diabetes mellitus, obesity, ischemia, and venous stasis (Handbook on Hyperbaric Medicine 2006; Menke et al. 2007).

Delayed wound healing is associated with factors such as older age, elevated levels of reactive oxygen species, inflammatory cytokines, and microbial biofilms (Caley et al. 2015; Oso et al. 2018; Thakur et al. 2011). For wound healing, various treatments like antibiotic ointments, hydrocortisone, and disinfectant solutions (e.g., acetic acid and betadine) are effective (Sewall et al. 2003). Additionally, some medicinal herbs are considered safe and effective alternatives for wound healing (Table 9, Fig. 2, online Appendix Fig. 9).

1. *Achillea millefolium* (Yarrow), has anti-inflammatory, antirheumatic, antiseptic, anti-prostaglandin, antiulcer, and antidiuretic effects due to its constituents like flavonoids, essential oil, sesquiterpene lactones, and dicaffeoylquinic acid. In vivo studies have shown that it reduces wound size and scar tissue thickness, aiding in wound healing (Jalali et al. 2010).
2. *Hypericum perforatum* (St. John's Wort), known for its anti-inflammatory properties, *Hypericum perforatum* is rich in hypericin and hyperforin. It promotes diabetic skin wound healing by enhancing collagen synthesis, proliferation, and revascularization (Yadollah-Damavandi et al. 2015).
3. *Calendula officinalis* (Pot Marigold), contains flavonoids, terpenoids, and phenolic acids with anti-inflammatory and antioxidant effects. It has been found to effectively heal wounds, promote collagen formation, and stimulate fibroblast activity, aiding in wound healing (Shafeie et al. 2015).
4. *Aloe barbadensis* (Aloe Vera), has antibacterial, anti-diabetic, and anti-inflammatory properties due to glycosides, acemannan, and polysaccharides. These components enhance T cell activity, phagocytosis, and collagen synthesis, accelerating wound healing (Xing et al. 2015).
5. *Alternanthera sessilis* and *Alternanthera brasiliana*, have antibacterial and cytotoxic activities. Chloroform extracts from *Alternanthera* leaves can significantly reduce wound area and promote re-epithelialization in animal models (Jalalpure et al. 2008). *Alternanthera brasiliana* accelerates the healing process in burn wounds by increasing the content of vitamin C, SOD, CAT, protein, GSH, and hydroxyproline in damaged tissue (Barua et al. 2012).
6. *Heliotropium indicum*, contains pyrrolizidine alkaloids and has traditionally been used for various ailments. Methanolic extract of *Heliotropium indicum* stimulates wound healing by increasing catalase and SOD activity, granulation tissue weight, and hydroxyproline content (Dash and Murthy 2011).
7. *Curcuma longa* (Turmeric) with Curcumin, its active compound, has anti-inflammatory, antioxidant, and anti-infectious effects. Curcumin promotes wound healing by deposition of collagen, tissue remodeling, and granulation tissue formation (Joe et al. 2010).
8. *Moringa olifera* (Drumstick), promotes wound healing by enhancing cell proliferation and migration of human dermal fibroblast cells (Gothai et al. 2016). It also reduces wound size and accelerates healing in diabetic-induced rats (Muhammad et al. 2016).
9. *Lantana camara* (Lantana), exhibits wound-healing properties by increasing collagen synthesis and wound contraction rate (Nayak et al. 2009). It contains essential oils with antifungal, antibacterial, and antimicrobial activity (Deena and Thoppil 2000).
10. *Centella asiatica* (Gotu Kola), with active glycosides Madecassoside and asiaticoside enhance collagen synthesis and wound contraction, promoting wound healing (Wu and Wang 2012b). It improves chronic ulcer healing in animal models (Somboonwong et al. 2012).
11. *Carica papaya* (Papaya) latex enhances re-epithelialization and hydroxyproline content in damaged tissue, aiding in burn wound healing (Gurung and Škalko-Basnet 2009). The stem of *Carica papaya* accelerates wound healing in albino rats (Singh et al. 2020).
12. *Hibiscus Rosa sinensis* (Shoeblackplant), enhances cellular proliferation and collagen synthesis at the wound site, aiding in wound healing (Bhaskar and Nithya 2012).
13. *Mimosa pudica*, has anti-inflammatory and antinociceptive properties due to compounds like beta-sitosterol and mimosine. Methanolic extracts of *Mimosa pudica* stimulate wound contraction and reduce wound area in animal models (Paul et al. 2010).

Table 9 Plant based interventions for Wounds

Plant (scientific name)	Bioactive compounds	Mechanism of action	Extract	Dose	Experimental model	Results
1. <i>Achillea millefolium</i>	Proazulenes, Dicafeoylquinic acid (Benedek et al. 2007, 2008)	Prevention of Blood loss by coagulation, Debridement of wound, tissue remodeling and collagen positioning	Alcoholic extract (Jalali et al. 2010)	5 ml/day (Jalali et al. 2010)	Rabbit (Jalali et al. 2010)	Accelerates burn-induced wound healing (Jalali et al. 2010)
2. <i>Hypericum perforatum</i>	Hypericin, hyperforin (Saddiqe et al. 2010; Wölflle et al. 2013)	Fibroblasts proliferation inducing effect	Ethanol extract (Yadollah-Damavandi et al. 2015)	5% HP gel	Rat (Yadollah-Damavandi et al. 2015)	Improves tissue regeneration (Yadollah-Damavandi et al. 2015)
3. <i>Calendula officinalis</i>	Kaempferol, quercetin, isorhamnetin (Re et al., 2009b)	Enhances antioxidant defence mechanism (Chandran and Kuttan 2008)	Ethanol extract (Chandran and Kuttan 2008)	200 mg/kg (Chandran and Kuttan 2008)	Rat (Chandran and Kuttan 2008)	Increases collagen hydroxyproline and hexosamine contents and improves wound healing (Chandran and Kuttan 2008)
4. <i>Aloe barbadensis</i>	Acemannan (Salehi et al. 2018)	Promotes T cell activity, phagocytosis and trans of IL-6, IL- α , TNF- α and IL-1 β etc. proinflammatory mRNAs (Ali et al. 2014)	Commercial Aloe vera gel	300 mg/kg mannose sugar (Ali et al. 2014)	Mouse (Davis et al. 1994)	improved wound healing process (Ali et al. 2014)
5. <i>Alternanthera sessilis</i> and <i>Alternanthera brasiliana</i>	Stigmasteol, beta-sitosterol, kaempferol, quercetin (Pereira et al. 2013)	HaCat cells & HDFs Proliferation and migration. Growth factor increasing such as KGF, PDGF, VEGF	Chloroform extract (Jalalpure et al. 2008)	200 mg/kg body weight (Jalalpure et al. 2008)	Rat (Jalalpure et al. 2008)	Reduces wound area and increases re-epithelialisation (Jalalpure et al. 2008)
6. <i>Heliotropium indicum</i>	Helindicine, lycopsamine, heleruine-N-oxide, europine-N-oxide (Dash and Abdullah 2013; Souza et al. 2005)	Increasing the formation of collagen & less macrophages	Methanolic extract (Dash and Murthy 2011)	(300, 600 and 2000) mg/kg (Dash and Murthy 2011)	Rat (Dash and Murthy 2011)	Increases SOD and catalase activity, tissue granulation weight and the hydroxyproline content (Dash and Murthy 2011)
7. <i>Curcuma longa</i>	Curcumin (Akbik et al. 2014)	Heals wounds through deposition of collagen, tissue remodeling and granulation tissue formation (Joe et al. 2010)	Ethanol extract (Pawar et al. 2015)	150 mg/kg body weight (Pawar et al. 2015)	Rat (Pawar et al. 2015)	Heals wound (Pawar et al. 2015)
8. <i>Moringa olifera</i>	4-isothiocyanate, S-methyl-N-thiocarbamate (Cheenpracha et al. 2010)	Increases cell proliferation and migration of HDF cell (Gothai et al. 2016)	Ethyl acetate extract (Gothai et al. 2016)	(12.5 and 25) microgram/ml (Gothai et al. 2016)	Human dermal fibroblast cells (Gothai et al. 2016)	Helps in wound healing (Gothai et al. 2016)
9. <i>Lantana camara</i>	Quinine (Venkatachalam et al. 2011)	10.5402/2011/847980 (Nayak et al. 2009)	Aqueous leaf extract (Nayak et al. 2009)	100 mg/kg/day (Nayak et al. 2009)	Rat (Nayak et al. 2009)	Increases collagen synthesis and wound contraction rate, decreases healing time (Nayak et al. 2009)
10. <i>Centella asiatica</i>	Madecassoside and asiaticoside (Chandrika and Prasad Kumara 2015; Moqbel et al. 2020)	Increased anti-oxidant levels (Jamil et al. 2007)	Alcoholic extract (Jamil et al. 2007)	100 mg/L of extract	Rat (Jamil et al. 2007)	Helps in wound healing (Jamil et al. 2007)
11. <i>Carica papaya</i>	Kaempferol, quercetin, caffeic acid (Saikia et al. 2015; Sangsoy et al. 2017; Santana et al. 2019)	Increases rate and extent of wound closure and increases hydroxyproline levels (Nayak et al. 2007)	Aqueous extract (Nayak et al. 2007)	100 mg/kg/day (Nayak et al. 2007)	Rat (Nayak et al. 2007)	Reduces wound area, enhances hydroxyproline content and weight of dry granulation tissue (Nayak et al. 2007)

Table 9 continued

Plant (scientific name)	Bioactive compounds	Mechanism of action	Extract	Dose	Experimental model	Results
12. <i>Hibiscus Rosa sinensis</i>	Quinine, anthraquinones (Kaur and Lakshmanan 2015; Kumari et al. 2015)	Increases levels of hexosamine and uronic acid (Bhaskar and Nithya 2012)	Flower ethanolic extract (Bhaskar and Nithya 2012)	5/10 g of extract in 100 g of ointment base (Bhaskar and Nithya 2012)	Rat (Bhaskar and Nithya 2012)	Enhances cellular proliferation and collagen synthesis at wound site, increases DNA, protein and collagen content, improves re-epithelialisation rate and wound contraction (Bhaskar and Nithya 2012)
13. <i>Mimosa pudica</i>	Beta-sitosterol, mimosine (Jha 2007)	Collagen progression,	Methanolic shoot extract and methanolic root extract	250 g and 250 g	Rat	Increase excisional wound contraction rate and promotes wound healing activity

Discussion

Pain is a condition that drives many patients to explore complementary therapies. Herbal medicines have gained interest as potential solutions for pain management. In this review we have delved into the effects of herbal treatments on pain relief and the underlying mechanisms at play.

Our review emphasizes the positive outcomes associated with herbal medicines in managing pain. Existing evidence suggests that these therapies may effectively alleviate pain by targeting two mechanisms; reducing inflammation and mitigating oxidative stress. These therapeutic effects can be attributed to their ability to regulate signaling pathways and modulate immune cell activity. Notably herbal remedies like turmeric (*Curcuma longa*) have shown anti inflammatory properties by inhibiting pro inflammatory cytokines such, as interleukin 1 β (IL 1 β) and tumor necrosis factor alpha (TNF α) (Ammon 2006; Chainani-Wu 2003; Grzanna et al. 2005; Han et al. 2018; Jurenka 2009; McKenna et al. 2001; Siddiqui 2011). Similarly frankincense (*Boswellia serrata*) has demonstrated its ability to inhibit the production of leukotrienes which are known mediators of inflammation (Ammon 2006).

Furthermore herbal treatments have also been found to exhibit antioxidant effects. For instance, Ginkgo biloba has been found to possess antioxidant

properties by neutralizing free radicals and shielding against oxidative stress (McKenna et al. 2001). Similarly, Hypericum perforatum (St. Johns wort) has displayed the ability to boost the expression of antioxidant enzymes like superoxide dismutase (SOD) thereby providing defense against harm (Sánchez-Reus et al. 2007).

One of the advantages of herbal medicines for managing pain is their generally favorable tolerance, minimal side effects, affordability and some level of effectiveness (Klepser and Klepser 1999). Additionally, incorporating treatments into pain management empowers patients by allowing them to actively participate in choosing therapies they believe will be most beneficial.

The increasing popularity of complementary medicine can be attributed in part to the growing emphasis on patient autonomy and the widespread availability of online information. Dissatisfaction with therapies often stemming from a lack of confidence among both practitioners and patients (Winter and Korzenik 2017) has also contributed to this trend. It is vital for healthcare providers to stay knowledgeable about the range of pain therapies available, to their patients regardless of the underlying cause.

However it is essential for healthcare professionals to have an understanding of the risks and limitations associated with the use of herbal remedies in clinical

settings. For example cannabis, a herbal treatment for pain management contains active compounds like delta 9 tetrahydrocannabinol (THC) and cannabidiol (CBD) which interact with the body's endocannabinoid system to provide relief from pain (Russo 2008). However the use of cannabis for pain management remains a subject of controversy due to its effects, potential for misuse and varying legal status in different countries. Turmeric despite being studied for its potential benefits faces challenges such as its relatively low curcumin content and limited ability to be absorbed by the body effectively. This may require doses for therapeutic effects (Corroon and Felice 2019; Prasad et al. 2014). Moreover recommending supplements like curcumin to specific patient groups such as children or pregnant and breastfeeding women raises concerns due to limited conclusive evidence regarding their benefits. Additionally there are variations in the status of herbal medicines across different countries making it challenging to ensure their quality, safety and effectiveness—especially in regions where these therapies are not regulated by organizations like the US Food and Drug Administration (FDA).

Another significant concern relates to interactions, between herbal remedies and prescription medications that could potentially lead to adverse effects. For instance St. Johns wort (SJW) is a known herbal supplement that people often use to address depression and anxiety. It is important to note that SJW can interact with painkillers. When someone takes SJW, it triggers the activity of two enzymes called P450 3A4 (CYP3A4) and P glycoprotein (P gp). These enzymes are responsible for breaking down and eliminating prescription drugs, including opioids. This interaction can potentially speed up the metabolism and elimination of opioids which might reduce their effectiveness in providing relief from pain in cases. On the side SJW might also slow down the metabolism and elimination of opioids in specific individuals increasing the risk of respiratory depression or other adverse effects (Delgoda and Westlake 2004; Gurley et al. 2005).

Furthermore one significant challenge with medicines is the absence of standardized guidelines for dosages. Unlike prescription medications herbal remedies are not subject to regulatory oversight. As a result there is variation in terms of product composition and quality across different brands or sources.

This lack of standardization can lead to inconsistencies in potency and efficacy when it comes to medicines making it difficult to achieve consistent therapeutic outcomes (Bone and Mills 2013; Tachjian et al. 2010). Both patients and healthcare practitioners often face challenges due to the absence of guidance on appropriate dosages, for these herbal remedies. This can result in situations where patients may not take enough dosage leading to inadequate pain relief or taking too much dosage causing adverse effects or toxicity concerns (Dowell et al. 2022).

Given the intricacies involved it is crucial for patients to have conversations with their healthcare providers when it comes to using herbal medicines. It is important that healthcare professionals stay updated on the research about herbal therapies and how they might interact with prescription medications (Xiong et al. 2022).

While herbal supplements have the potential to improve the quality of life for patients dealing with chronic pain there is a concern that some individuals might prioritize these therapies over conventional medicine instead of using them as complementary treatments. This approach could result in pain management and potentially harmful outcomes (Cohen et al. 2021; Gardiner et al. 2008; Jermini et al. 2019). Additionally cost can be an obstacle since insurance coverage for herbal remedies is often limited or nonexistent. As a result patients are left to navigate the market of supplements on their own increasing the risk of misuse. The lack of regulation and quality control in the industry further compounds these challenges making it difficult for patients to make decisions about which products to choose (Shaw et al. 2012).

One notable issue highlighted is the awareness among healthcare providers regarding the use of herbal supplements, in clinical practice. Many patients report that they learn about alternative therapies from sources like the Internet, friends or naturopaths rather than from their healthcare providers (Nguyen et al. 2016). It is worth noting that medical schools have historically not dedicated attention to alternative medicine and herbal therapy in their curricula. As a result healthcare professionals may not feel discussing or advising on herbal supplements. To bridge this knowledge gap academic institutions should consider incorporating training on alternative medicine into their curricula in the future.

In our review of pain management approaches we emphasize the benefits of herbal therapies. These therapies are believed to offer pain relief by reducing inflammation and modulating stress. However it is important to evaluate these findings in the context of each study's methodology. Our assessment reveals a range of research methods used including randomized controlled trials (RCTs) and observational studies. Each approach has its strengths and limitations that contribute to the overall evidence base. Significantly the sample sizes varied greatly among these studies. This highlights the need for robust research, with adequately sized samples to validate these initial findings. Furthermore the diversity observed in preparations dosing and outcome measures makes it challenging to synthesize the results. This indicates the need for standardization in research on herbal medicine. Selection and performance bias was noticeable in various forms. These biases have the potential to skew the results.

When applying these findings to practice, healthcare professionals should carefully consider efficacy and safety aspects. It is crucial to prioritize safety and follow evidence based practices when considering herbal therapies. In conclusion while herbal therapies show promise for pain management due to their inflammatory and antioxidant properties.

However, it is important to exercise caution when incorporating remedies into clinical practice taking into account the potential risks, limitations and interactions with prescription drugs. Open and honest conversations between patients and healthcare professionals are crucial to ensure effective pain management. Continuous research endeavors are necessary to address existing knowledge gaps and insurance providers may consider covering treatments if there is substantial evidence supporting their efficacy. Given the intricacies involved in this matter individuals need to be well informed, before making any decisions.

Conclusion

Pain is a common reason for seeking medical attention, and our review has identified several promising herbal therapies for pain management based on the available evidence. These medicinal herbs, containing bioactive substances, offer pain

relief, anti-inflammatory, and antibacterial effects without harmful side effects. While herbal extracts hold potential as a safe and effective alternative for pain management, it's crucial to acknowledge their potential risks and limitations. Further research is essential to fully understand their mechanisms of action, determine efficacy, and optimize clinical use. Encouraging open discussions between patients and healthcare providers regarding the use of herbal supplements is a vital step toward improving pain management strategies.

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Data availability Dataset used in this study will be available as per request from the authors.

Declarations

Conflict of interest The authors report no competing interests. The authors alone are responsible for the content and writing of this article.

Ethical approval and consent to participate The review adheres to PRISMA-ScR (Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews) Ethical approval was not required for the conduct of this scoping review as it involves the synthesis of published literature and does not include any primary data collection from humans or animals.

Consent for publication Not applicable.

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